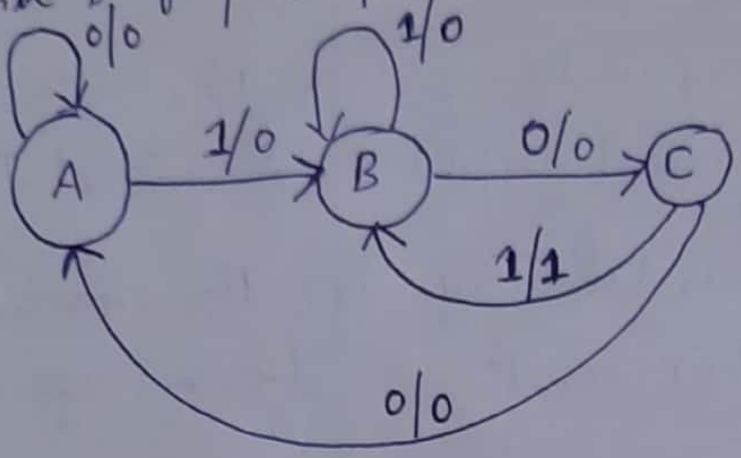
Write the difference between Moore & Mealy Machine

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dept of

Mealy Machine	Moore Machine (1)
* The output is a function of present state & present input.	* The output is a function of present state only.
* The change in the input affects the output of the circuit	* The change in the input does not affect the output.
* it requires lesser number of states (If the input is 3 bits, it requires 3 states to represent it.	* it requires more number of states. (If the input is 3 bits, it requires 4 states to represent it.
* it reacts faster to the input (ie., in the same clock cycle)	* it reacts slower to the input (ie., in the next clock cycle)
* it requires lesser hardware to implement the function.	* it requires larger hardware to implement the function.
* output is placed on transition.	* The output is placed on states.
* Asynchronous output generation.	* Synchronous output & State generation
* Not easier to design the circuit	* It is easier to design the circuit
* A Counter is not a Mealy Machine	* A Counter is a Moore Machine.

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1) Design a sequence detector 101 using Mealy Machine with D flip-flop.



State Table

present state	I/p	Next state	o/p
A	0	A	0
A	1	B	0
B	0	C	0
B	1	B	0
C	0	A	0
C	1	B	1

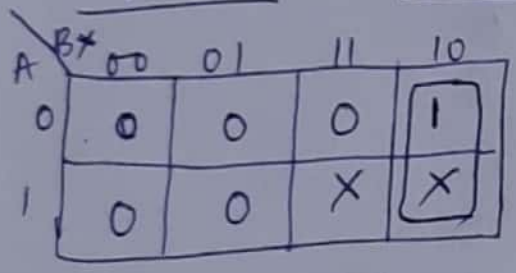
Assign A - 00, B - 01, C - 10

PS	I/p	N.S	o/p
AB		$A^+ B^+$	Z
00	0	00	0
00	1	01	0
01	0	10	0
01	1	01	0
10	0	00	0
10	1	01	1

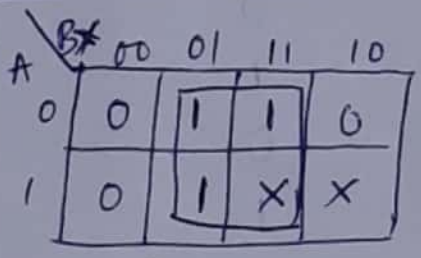
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P.S	I/P	N.S	O/P	F F I/P	
A B	X	A+ B+	Z	D_A	D_B
0 0	0	0 0	0	0	0
0 0	1	0 1	0	0	1
0 1	0	1 0	0	1	0
0 1	1	0 1	0	0	1
1 0	0	0 0	0	0	0
1 0	1	0 1	1	0	1

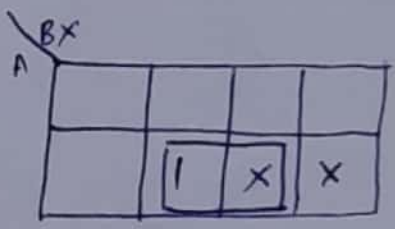
Designing using D flip-flop



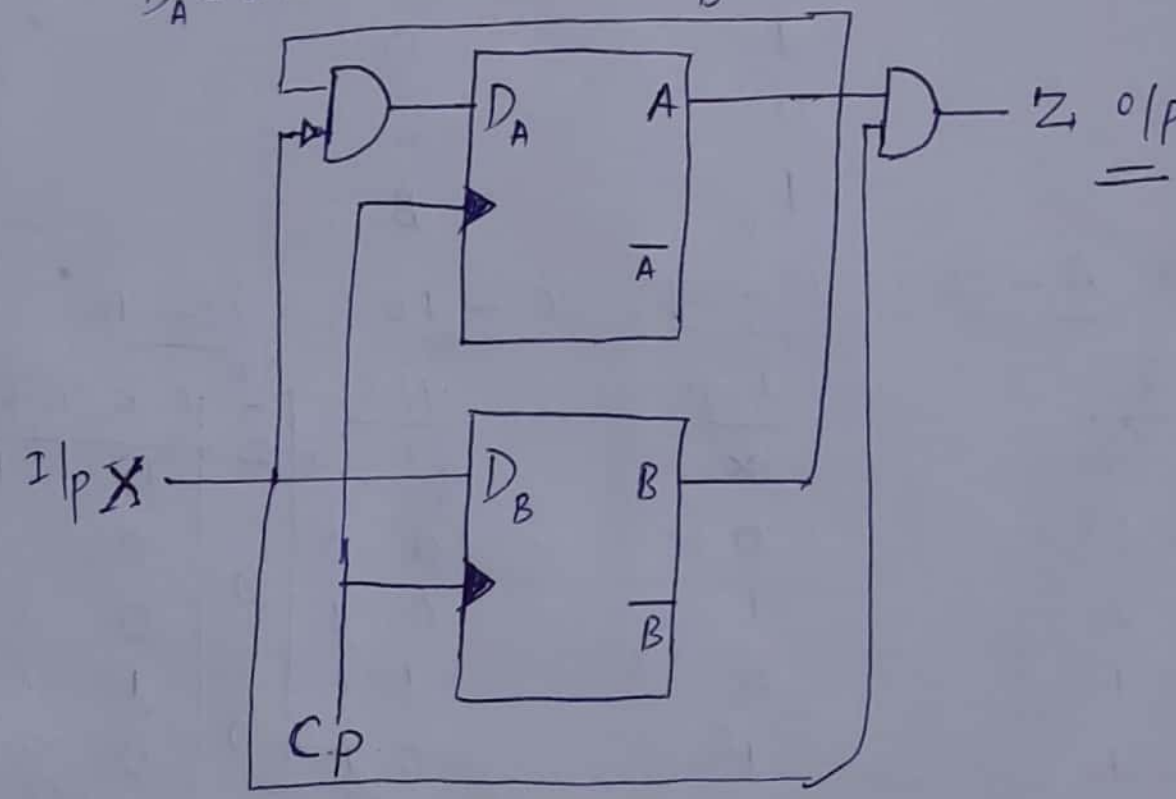
$D_A = B\bar{x}$



$D_B = x$

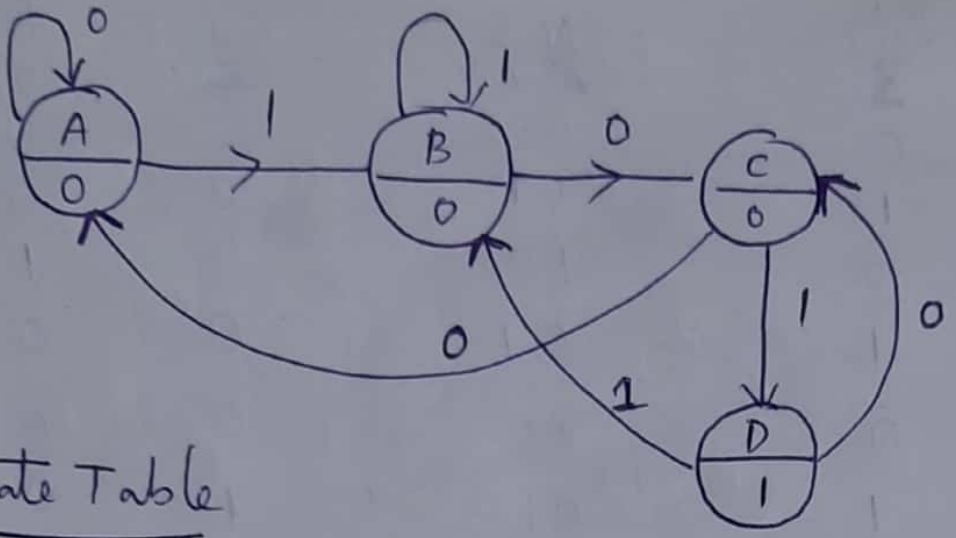


$Z = Ax$



2) Design a sequence detector to detect 101 using Moore Model.

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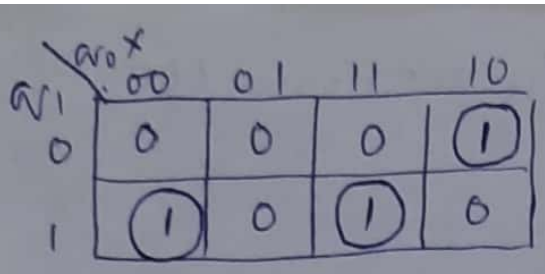
State Table

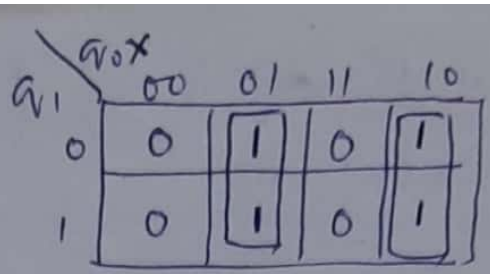
P S	I/P	N S
A	0	A
A	1	B
B	0	C
B	1	B
C	0	A
C	1	D
D	0	C
D	1	B

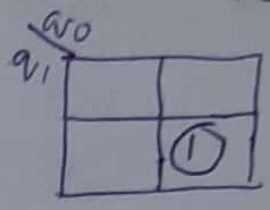
Assign A – 00, B – 01, C – 10, D – 11

P.S		I/P	N S		o/p	F F inputs	
q_1	q_0	X	q_1^+	q_0^+	Z	T_A	T_B
0	0	0	0	0	0	0	0
0	0	1	0	1		0	1
0	1	0	1	0	0	1	1
0	1	1	0	1		0	0
1	0	0	0	0	0	1	0
1	0	1	1	1		0	1
1	1	0	1	0	1	D	1
1	1	1	0	1		1	0

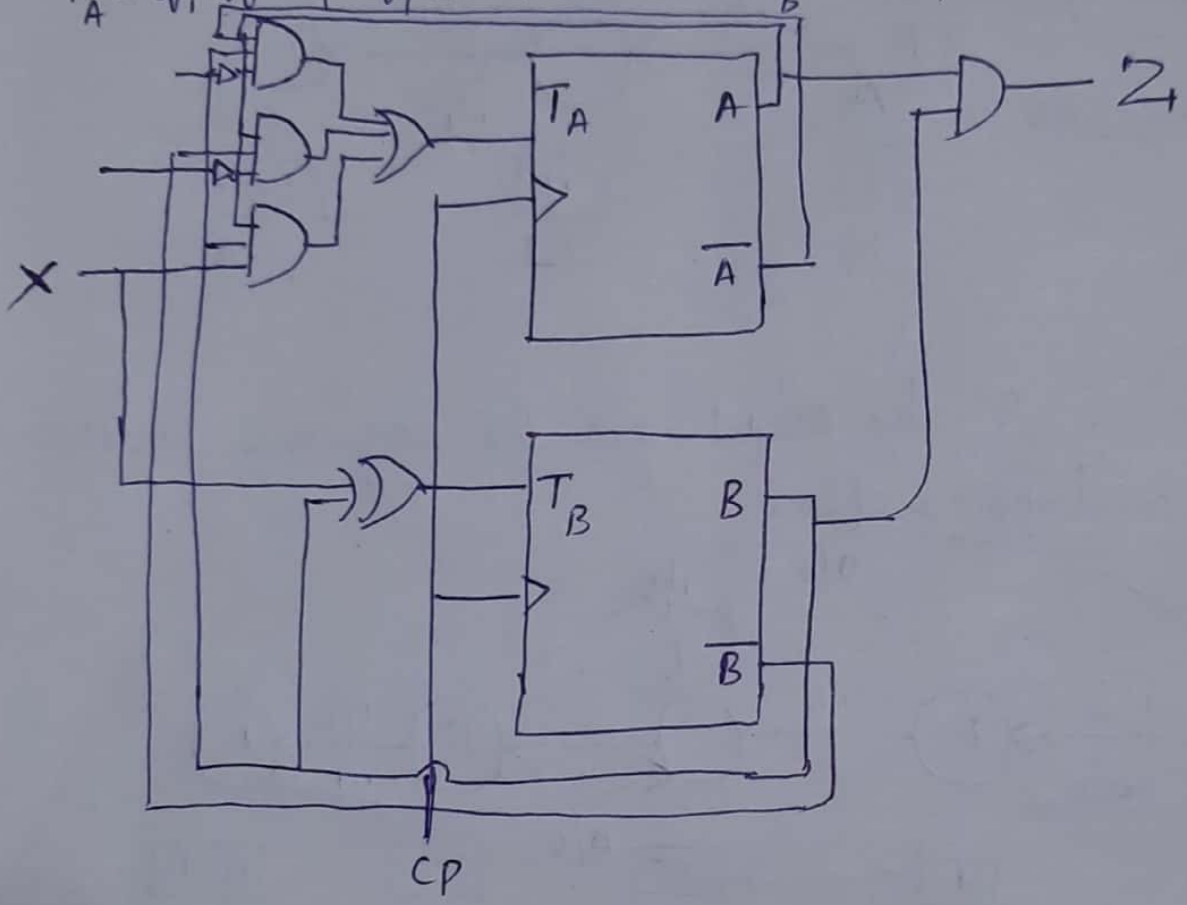
Scanned by CamScanner



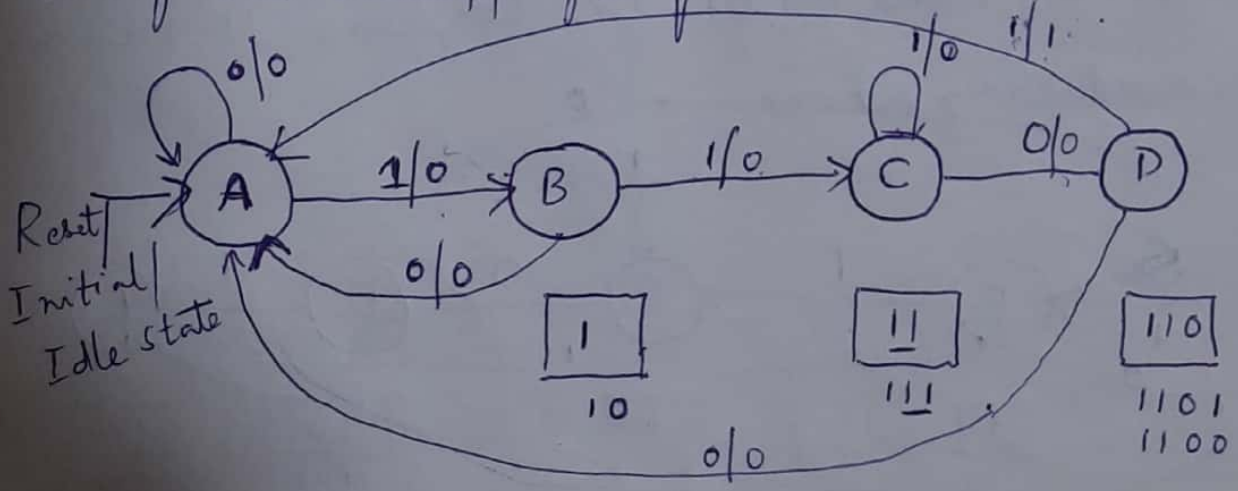




$T_A = \bar{q}_1 q_0 \bar{x} + q_1 \bar{q}_0 \bar{x} + q_1 q_0 x$ $T_B = \bar{q}_0 x + q_0 \bar{x}$ $Z = q_1 q_0$ (3)

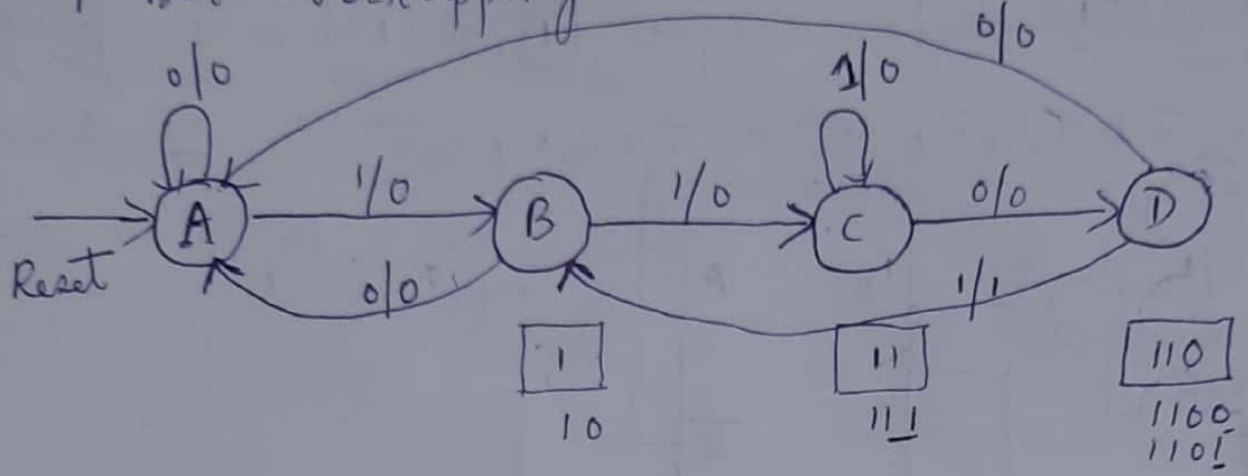


3) Design a Mealy model to detect the sequence 1101 Using non overlapping sequence

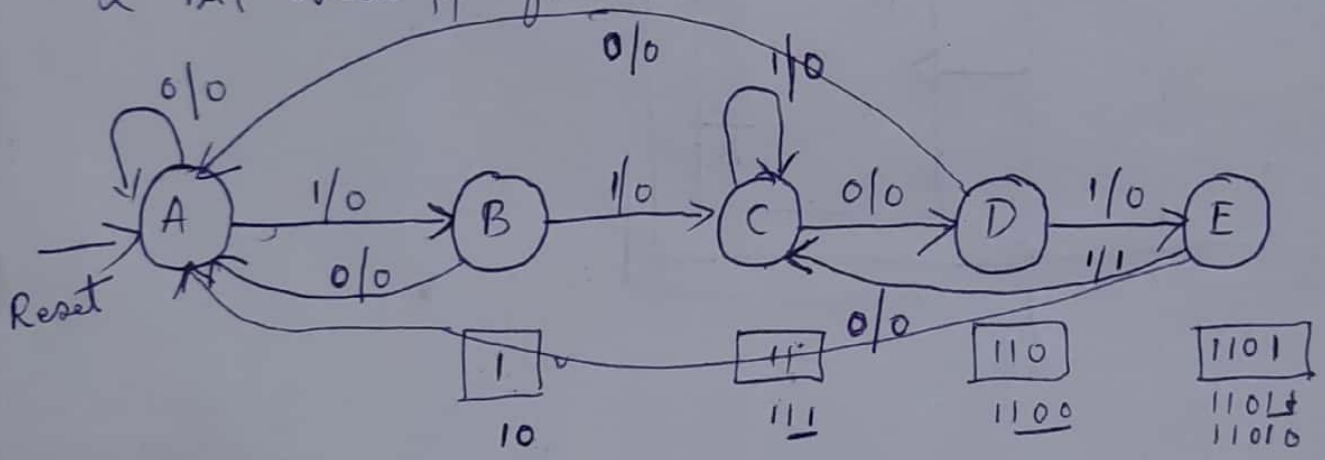


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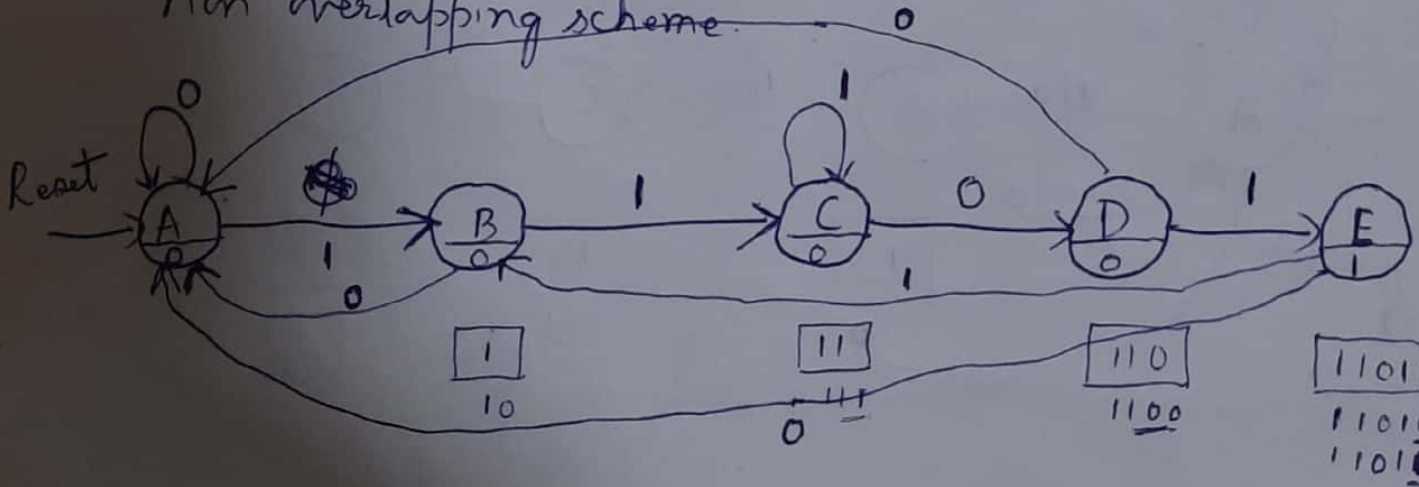
4) 16 Design a Meady Model for the sequence 1101 using 1 bit overlapping.



5) Design a Meady Model for the sequence 11011 using 2 bit overlappingscheme.

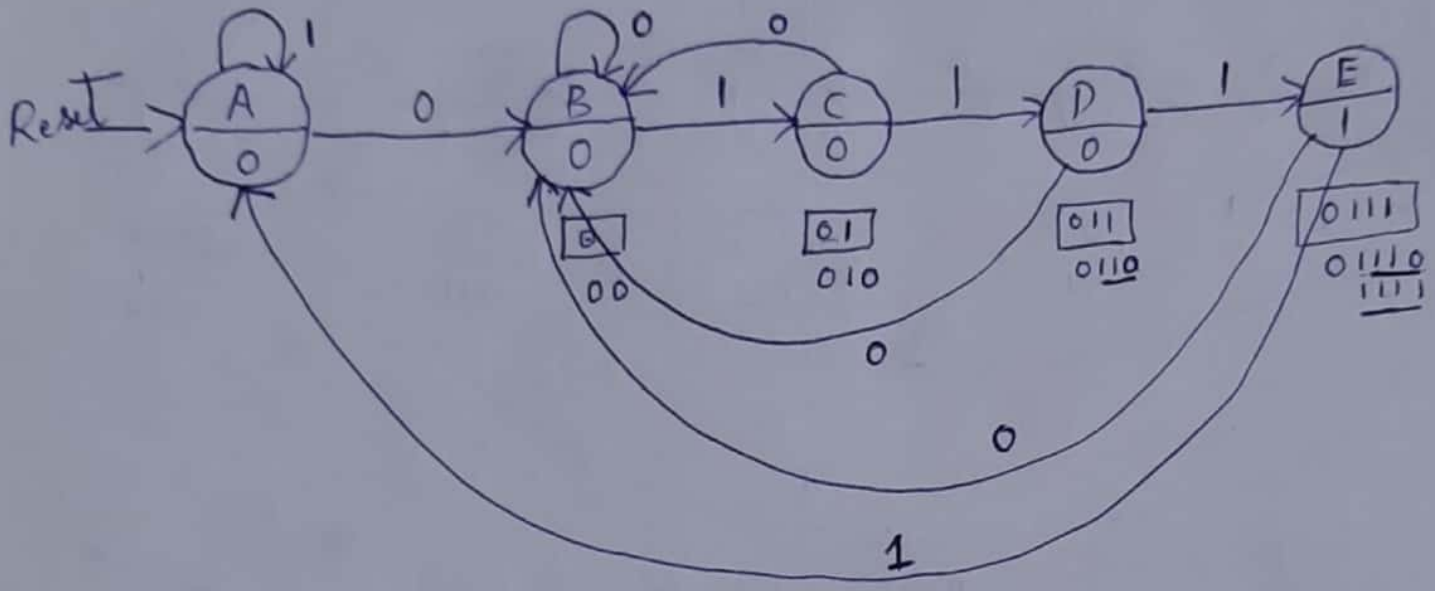


6) Design a Moore Model for sequence 1101 using non overlapping scheme.



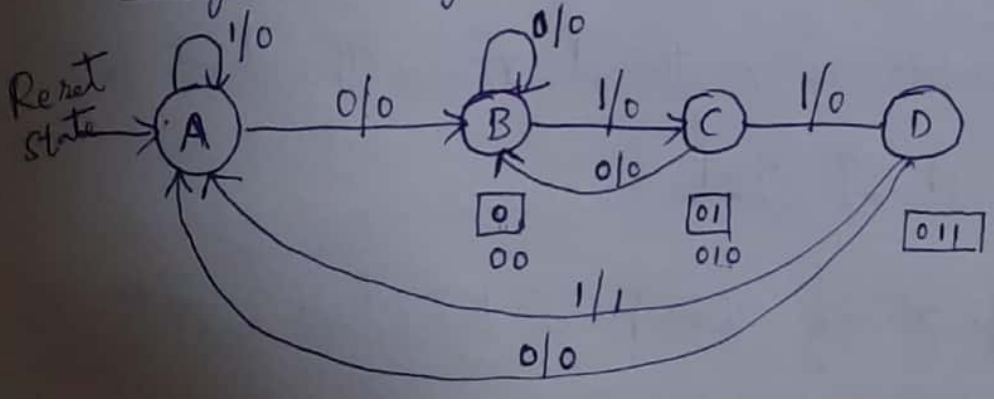
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Design a sequence detector to detect the sequence using D flip-flop. The input sequence is 0111 and (4) model is Moore Machine.



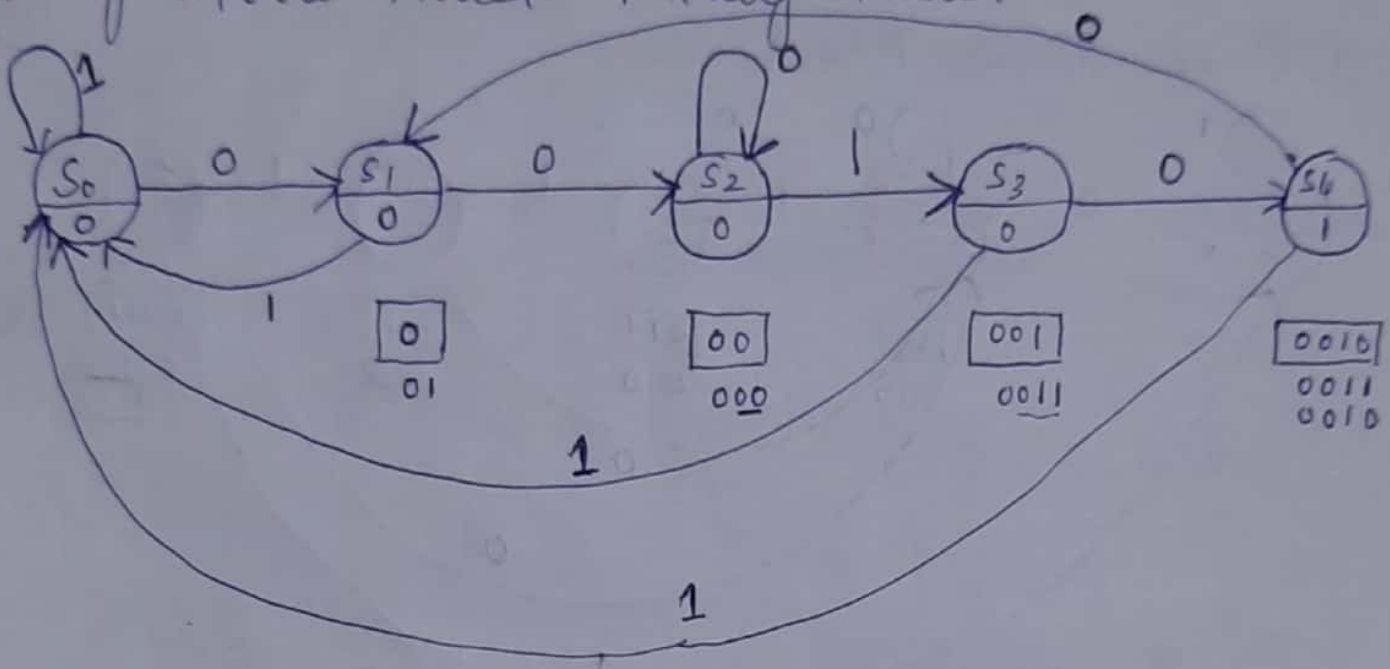
P.S	N.S		O/P
A B C	X=0	X=1	Z
A (0 0 0)	B	A	0
B (0 0 1)	B	C	0
C (0 1 0)	B	D	0
D (0 1 1)	B	E	0
E (1 0 0)	B	A	1

Design a sequence detector to detect the sequence 0111 using mealy model.

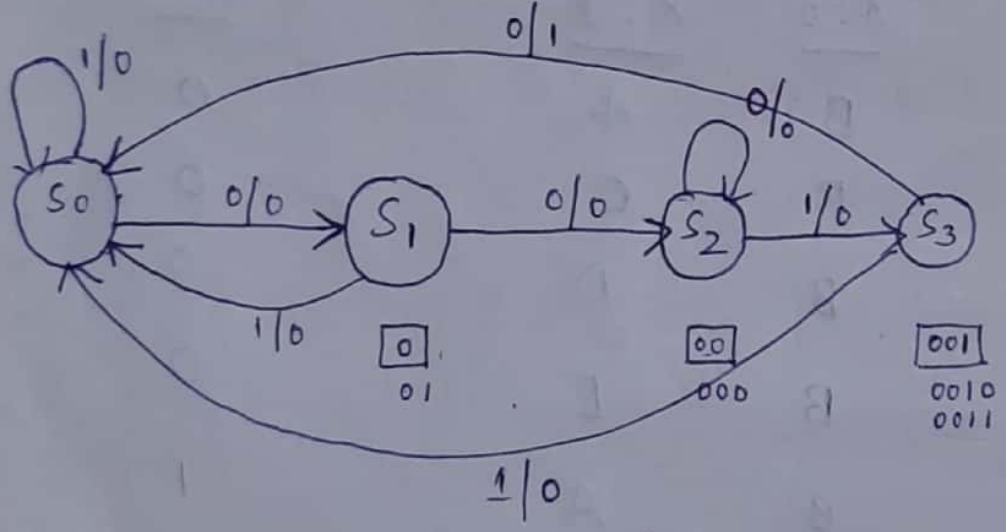


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34 Design a sequence detector to detect the sequence 0010 using Moore model & Mealy model.



Moore Machine



Mealy Machine

In Mealy Model

* For non overlapping method, move the last bit to the reset / Initial state.
* For 1 bit overlapping, Compare the last bit to

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the 1 bit state

* For 2 bit overlapping method, Compare the last two bits to the two bit state if not Compare 1 bit to the 1 bit state and so on.

In Moore Model

* For non overlapping scheme, Compare the last bit to the 1 bit state.

* For 1 bit overlapping scheme, Compare the last two bits to 2 bit state, then 1 bit state and so on.

* For 2 bit overlapping scheme, Compare the last 3 bits to 3 bits state, then 2 bit states and so on.

Default :- In Mealy model, the default is 1 bit overlapping.

In Moore model, the default is non overlapping scheme, Since it requires more number of states Compared to mealy model.

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08 Write the difference between Synchronous and Asynchronous Counter

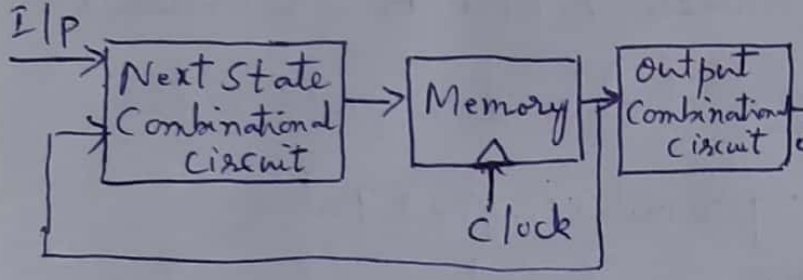
Asynchronous	Synchronous
In this type of Counter, flip-flops are Connected in such a way that the output of the first flip drives the clock for the next flip-flop.	In this type of Counters, There is no Connection between output of first flip-flop and clock input of the next flip-flop.
* All the flip-flops are not clocked simultaneously.	* All the flip-flops are clocked simultaneously.
* Logic circuit is very simple even for more number of states.	* Design involves complex logic circuit as number of states increases.
* The main drawback of these Counters is their low speed as the clock is propagated through number of flip-flops before it reaches last flip-flop.	As clock is simultaneously given to all flip-flops, there is no problem of propagation delay. Hence they are preffered when number of flip-flops increases in the given design.

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Difference between Moore & Mealy Machine (6)

Moore Machine

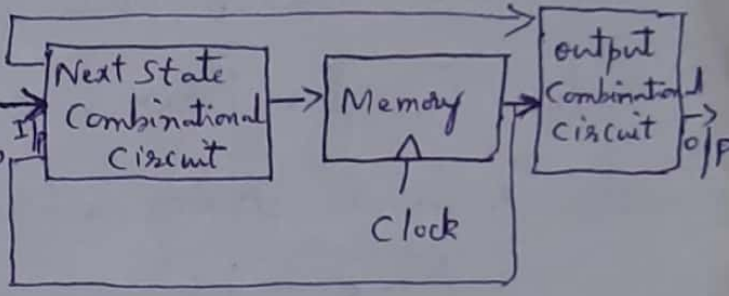
* The output is a function of present state.



* Generally it has more states than mealy m/c {(n+1) states}
* Change in input does not affect the output.
* The value of the o/p function is a function of transition and changes when the input logic on the present state.
* it reacts slower to the input.

Mealy Machine

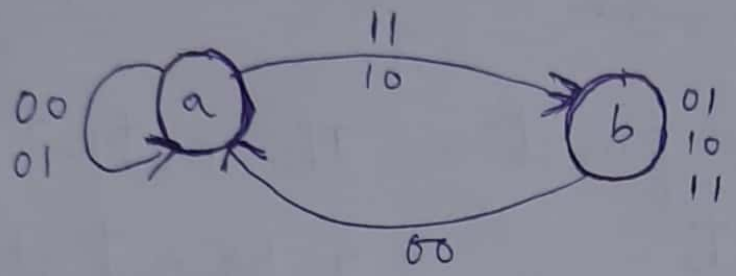
The output is a function of present state & input.



* Generally it has lesser states compared to moore machine {(n) states}.
* Change in the input will affect the output.
* The value of the output function of the function of current state & changes at the clock edges.
* it reacts faster to the input

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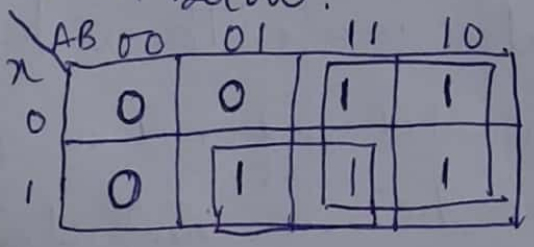
1) Design an asynchronous sequential circuit for the state transition diagram.



Let the two input variables A and B is Considered in order, Then the primitive flow table for given state diagram is shown below: primitive flow Table

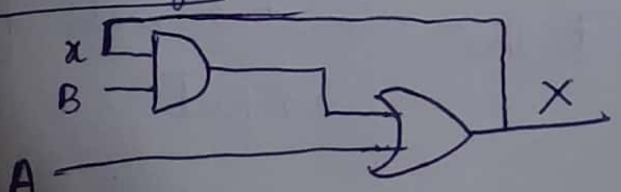
P.S	N.S			
	00	01	10	11
A	(a)	(a)	b	b
B	a	(b)	(b)	(b)

Let the output variable be X, then we assign state a as $x=0$ and b as $x=1$. Therefore, we can derive Equation as shown below:



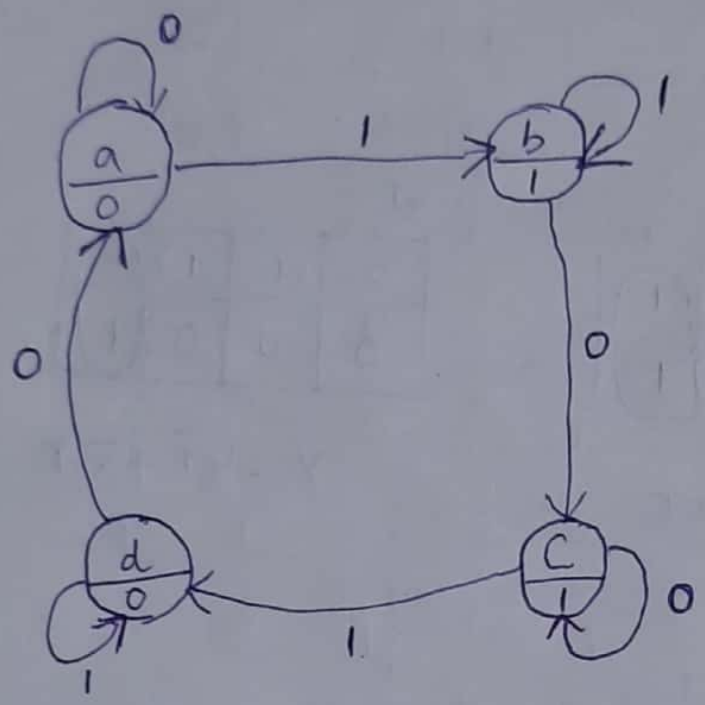
$$X = A + xB$$

The Circuit diagram



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Design a parity generator using asynchronous sequential logic that gives output = 1, when it receives odd number of pulses and output = 0, if the number of pulses received is even. (7)



primitive flow Table

PS	NS T=0	NS T=1	o/p Q
a	a	b	0
b	c	b	1
c	c	d	1
d	a	d	0

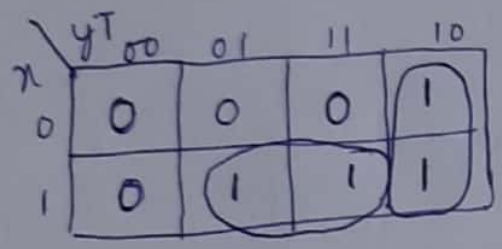
Assign a = 00, b = 01, c = 11 and d = 10, we can derive the design equation.

Assign present state is x y and Next state x+ & y+

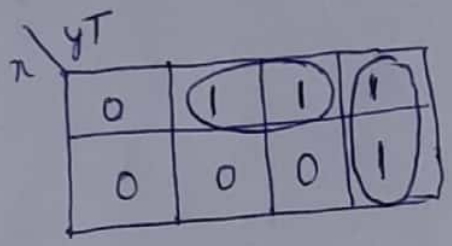
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12

P.S	Next state $x^+ y^+$		o/p
$x\ y$	$T=0$	$T=1$	Q
0 0	0 0	0 1	0
0 1	1 1	0 1	1
1 1	1 1	1 0	1
1 0	0 0	1 0	0



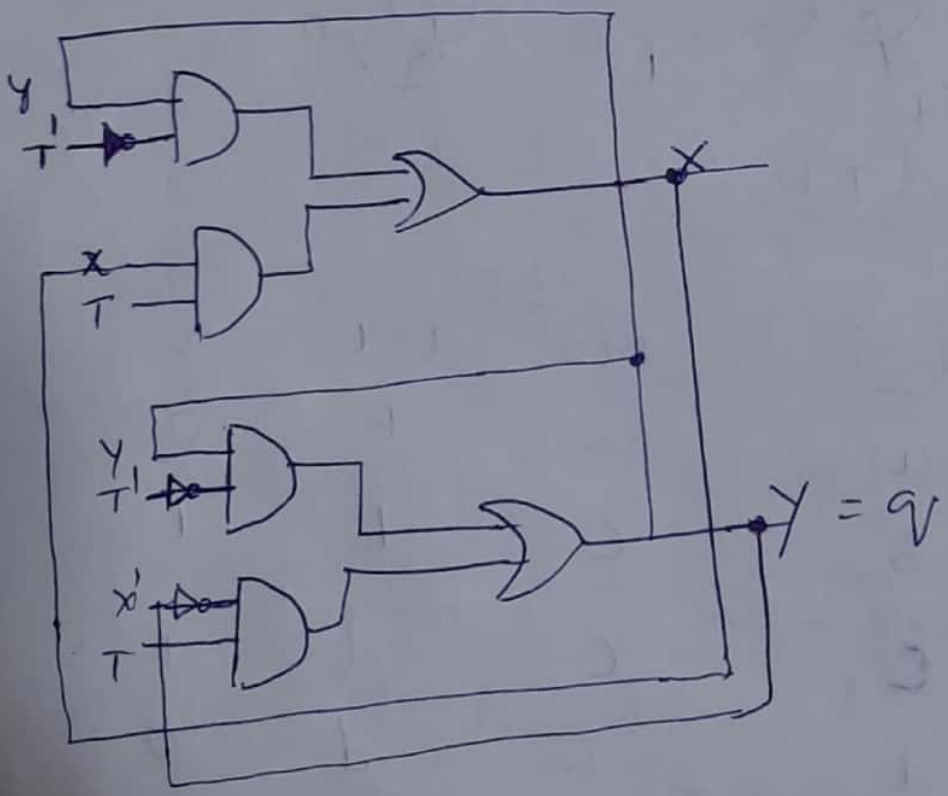
$X = y\bar{T} + xT$



$Y = y\bar{T} + \bar{x}T$



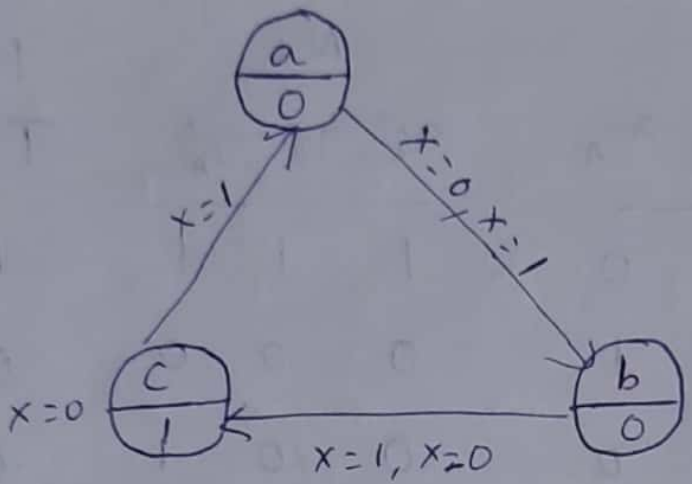
$Q = y$



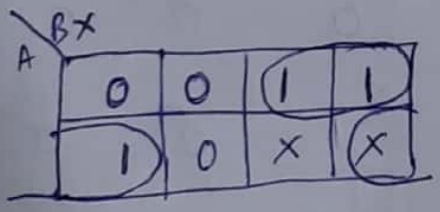
The circuit diagram

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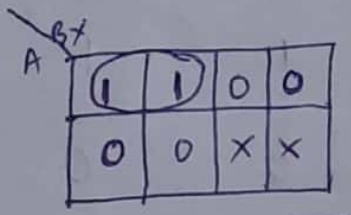
Realize the sequential circuit for the state diagram shown below: (8)



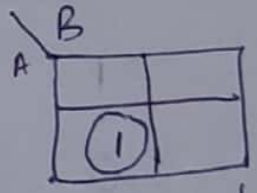
	P.S	N.S x=0		N.S x=1		O/P
	AB	A^+	B^+	A^+	B^+	Z
a –	00	0	1	0	1	0
b –	01	1	0	1	0	0
c –	10	1	0	0	0	1



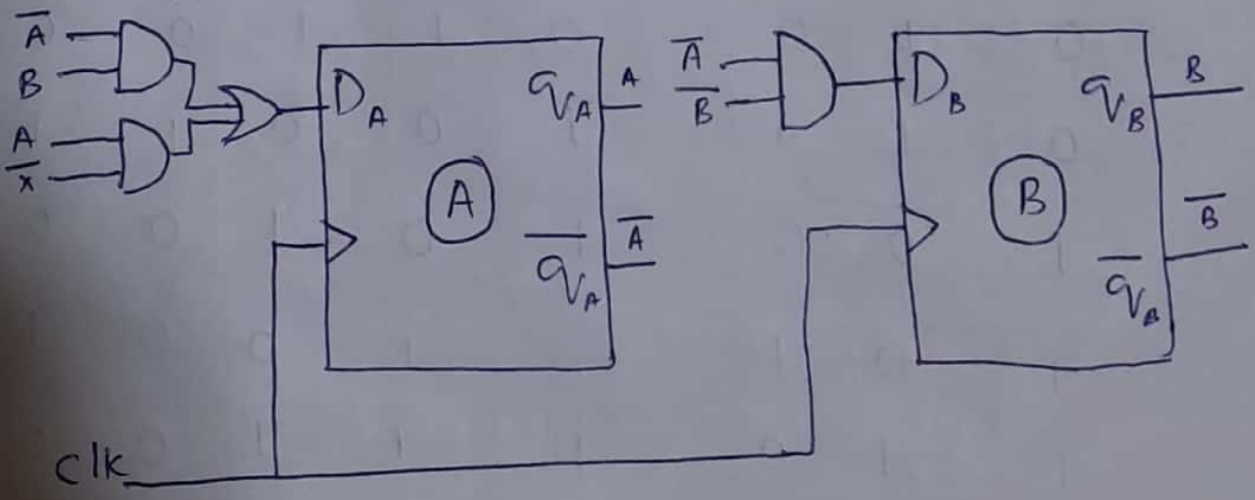
$D_A = \bar{A}B + A\bar{x}$



$D_B = \bar{A}\bar{B}$



$Y = AB'$

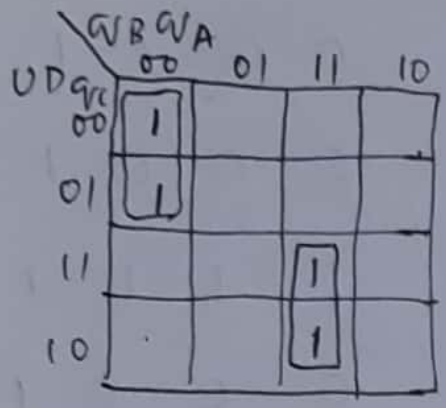


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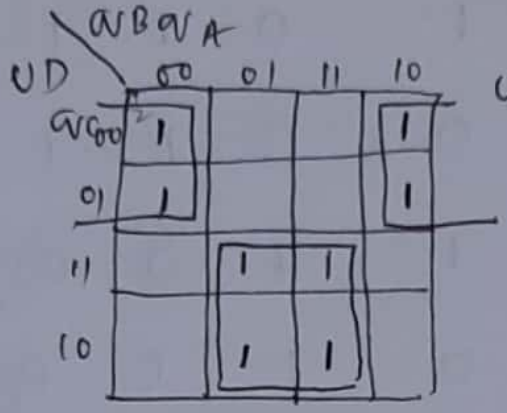
14 Design of up/down Synchronous Counter using T flip-flop.

I/P	P.S			N.S			FF I/P		
up/down (UD)	Q_C	Q_B	Q_A	Q_C^+	Q_B^+	Q_A^+	T_C	T_B	T_A
0	0	0	0	1	1	1	1	1	1
0	0	0	1	0	0	0	0	0	1
0	0	1	0	0	0	1	0	1	1
0	0	1	1	0	1	0	0	0	1
0	1	0	0	0	1	1	1	1	1
0	1	0	1	1	0	0	0	0	1
0	1	1	0	1	0	1	0	1	1
0	1	1	1	1	1	0	0	0	1
1	0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	0	1	1
1	0	1	0	0	1	1	0	0	1
1	0	1	1	1	0	0	1	1	1
1	1	0	0	1	0	1	0	0	1
1	1	0	1	1	1	0	0	1	1
1	1	1	0	1	1	1	0	0	1
1	1	1	1	0	0	0	1	1	1

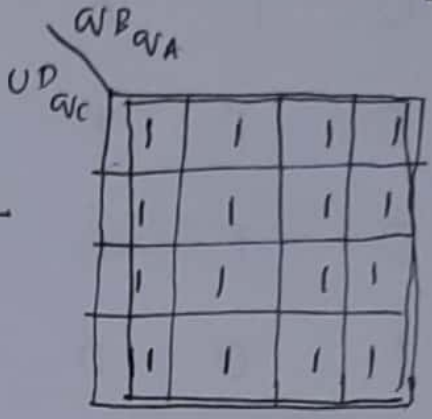
Scanned by CamScanner



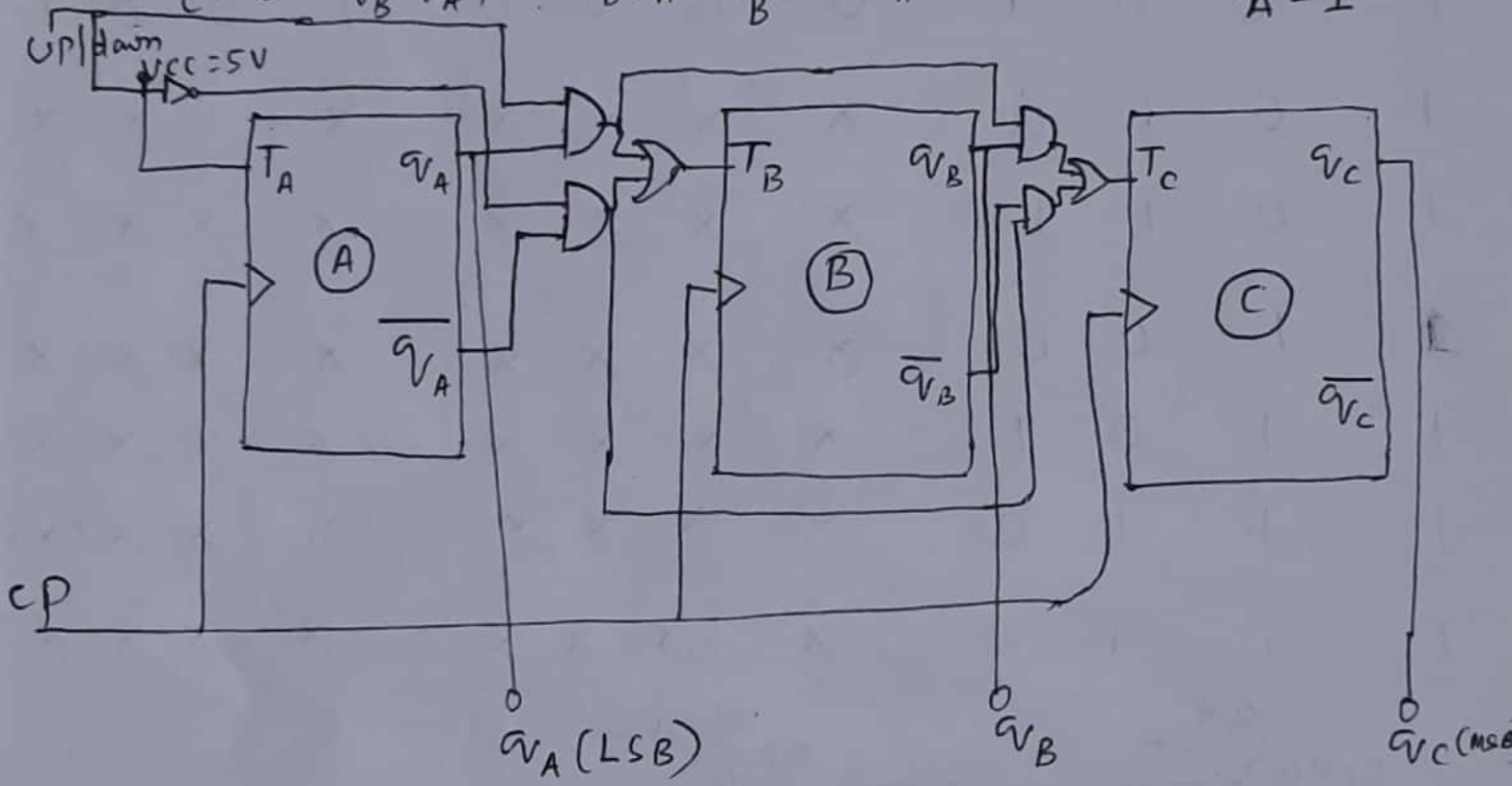
$T_C = \overline{UD}\,\overline{Q_B}\,\overline{Q_A} + UD\,Q_B Q_A$



$T_B = \overline{UD}\,\overline{Q_A} + UD\,Q_A$



$T_A = 1$



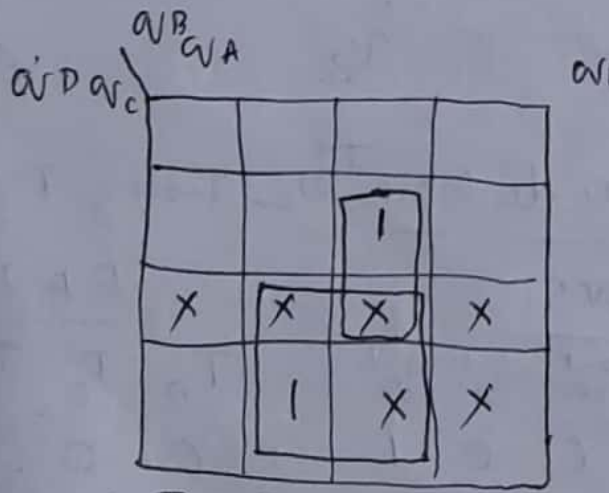
Design of Synchronous Decade Counter using T flip-flop

P.S				NS				FF I/p			
Q_D	Q_C	Q_B	Q_A	Q_D^+	Q_C^+	Q_B^+	Q_A^+	T_D	T_C	T_B	T_A
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1

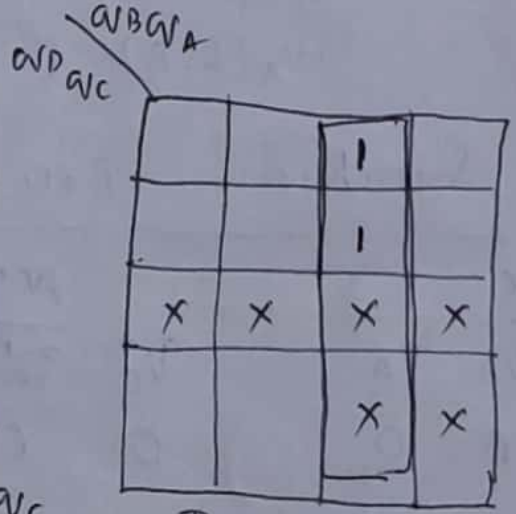
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28

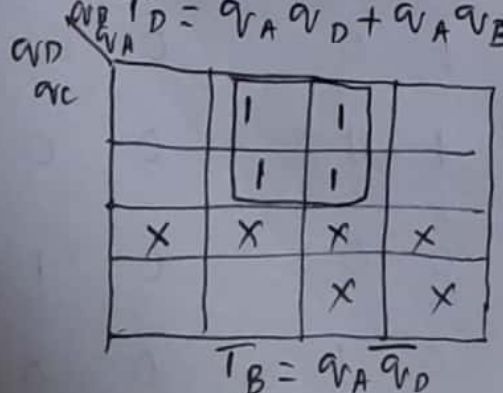
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1
1	0	1	0	X	X	X	X	X	X	X	X
1	0	1	1	X	X	X	X	X	X	X	X
1	1	0	0	X	X	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X	X



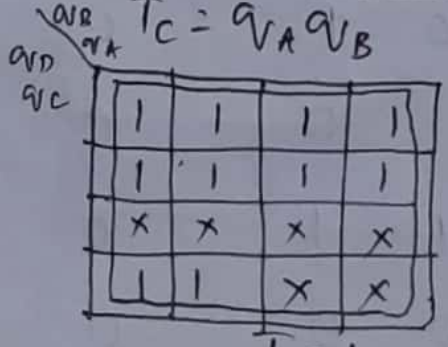
$T_D = q_A q_D + q_A q_B q_C$



$T_C = q_A q_B$

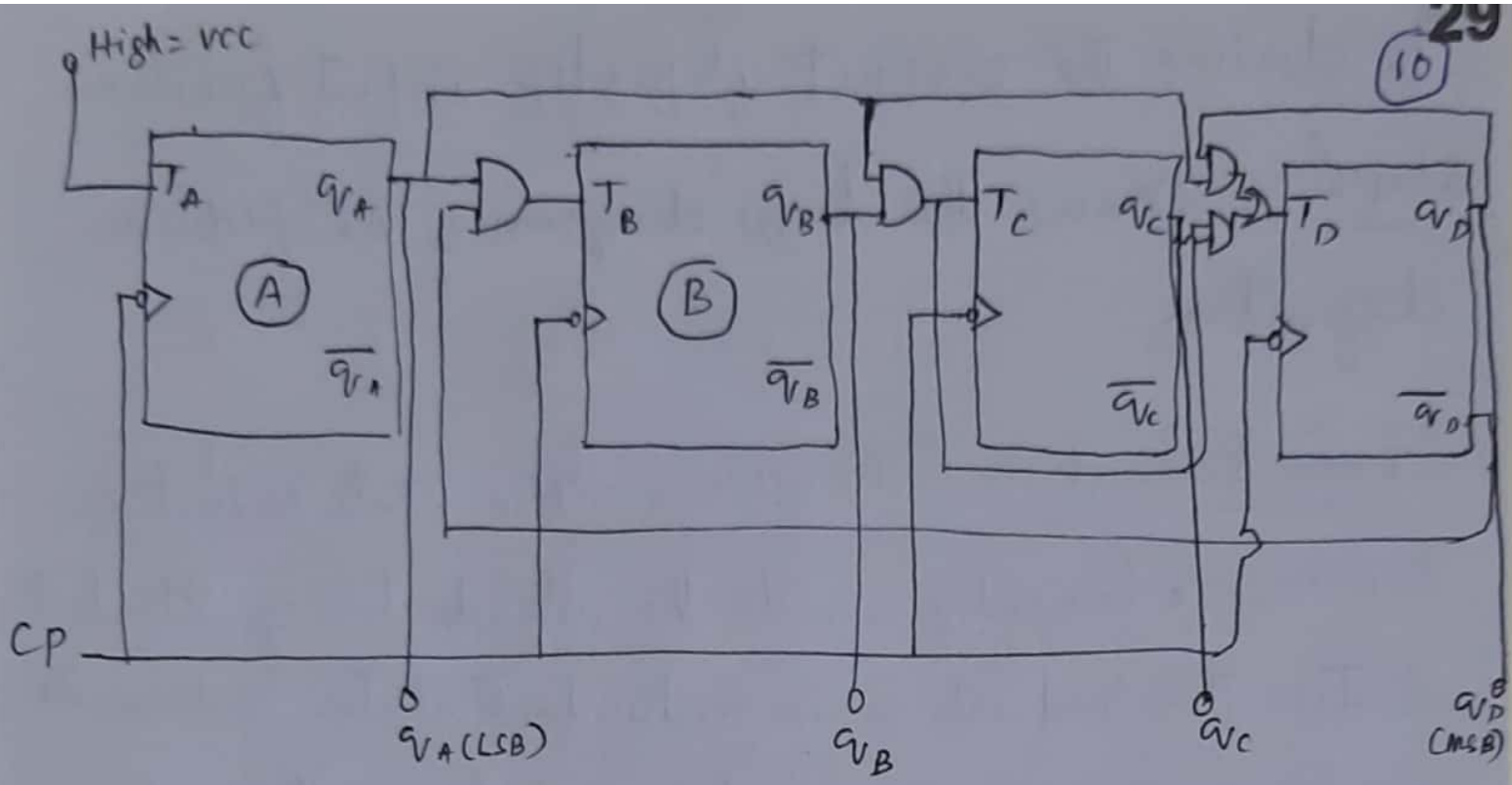


$T_B = q_A \overline{q_D}$



$T_A = 1$

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Design of Synchronous Counters (Counter design as a Synthesis problem)

Step 1 :- obtain the state transition diagram from the given circuit information (problem definition).

Step 2 :- Determine the number of flip-flops needed.
i.e., $2^N \geq S$

Step 3 :- Choose the type of flip-flops to be used.

Step 4 :- From the transition table, derive the circuit excitation table.

Step 5 :- Use k-map or any other simplification method

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to derive the circuit flip-flop input function.

Steps :- Draw the logic diagram of the problem definition.

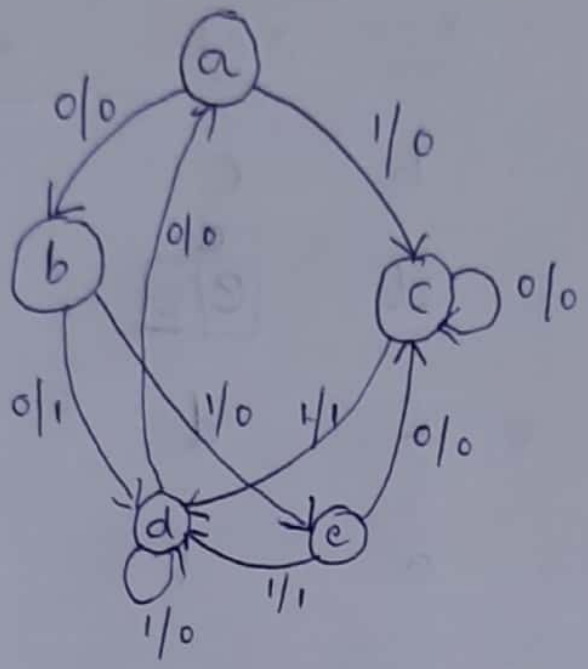
State Reduction Technique :- The state reduction technique basically avoids the introduction of redundant states. The reduction in redundant states reduces the number of required flip-flops and logic gates, reducing the cost of the final circuit.

1. Raw Elimination method :- In this method, the two states are said to be redundant or equivalent, if every possible set of inputs generate exactly same output & same next state. When two states are Equivalent, one of them can be removed without altering the input-output relationship.

1) Reduce the number of states for the given state diagram by Raw elimination method.

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(11)



Step 1 :- Determine the state table for given state diagram.

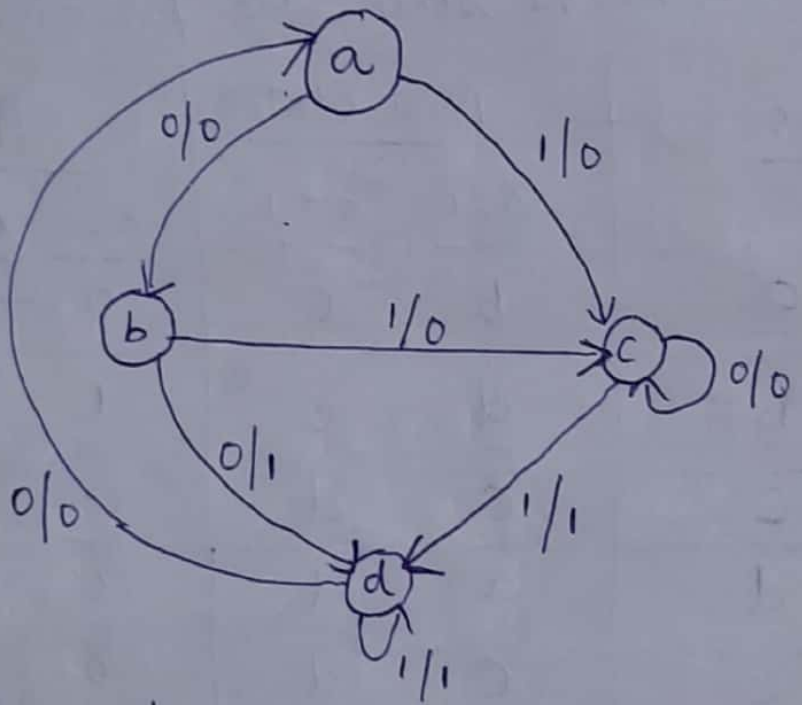
P.S	Next state		O/P	
	X=0	X=1	X=0	X=1
a	b	c	0	0
b	d	e	1	0
c	c	d	0	1
d	a	d	0	0
e	c	d	0	1

Step 2 :- Find Equivalent states :- States c & e are equivalent. Hence one state can be removed. State e is removed and replaced by c.

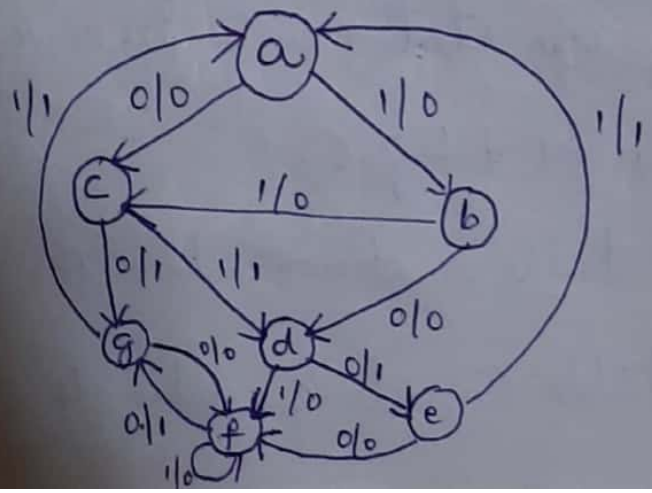
The final reduced table is shown below. The state diagram for the reduced table consists of only 4 states & shown below.

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P.S	N.S x=0	N.S x=1	O/P (Z) x=0	O/P (Z) x=1
a	b	c	0	0
b	d	c / e / c	1	0
c	c	d	0	1
d	a	d	0	0



2) Design the reduced state diagram for the given problem below by Raw Elimination method



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Step 1 :- Determine the state table for given state diagram (12)

P.S	N.S x=0	N.S x=1	output(z) x=0	output(z) x=1
a	c	b	0	0
b	d	c	0	0
c	g	d	1	1
d	e	f	1	0
e	f	a	0	1
f	g	f	1	0
g	f	a	0	1

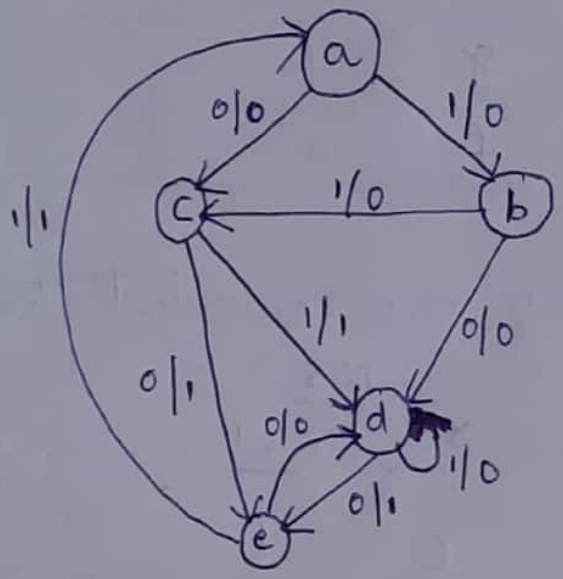
Step2 :- States e & g are Equivalent. Replace g by e.

P.S	N.S x=0	N.S x=1	o/p(z) x=0	o/p(z) x=1
a	c	b	0	0
b	d	c	0	0
c	e	d	1	1
d	e	f	1	0
e	f	a	0	1
f	e	f	1	0

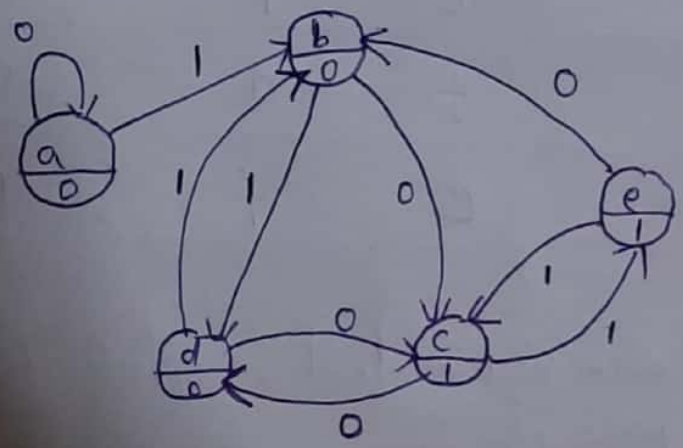
States d & f are Equivalent, replace f by d & the reduced state table is :

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P.S	N.S		O/P(Z)	
	X=0	X=1	X=0	X=1
a	c	b	0	0
b	d	c	0	0
c	e	d	1	1
d	e	d	1	0
e	d	a	0	1



3) Reduce the given state transition diagram by Row Elimination method.



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Step1 :- Represent the state table for the state diagram:

P.S	N.S X=0	N.S X=1	O/P (Z)
a	a	b	0
b	c	d	0
c	d	e	1
d	c	b	0
e	b	c	1

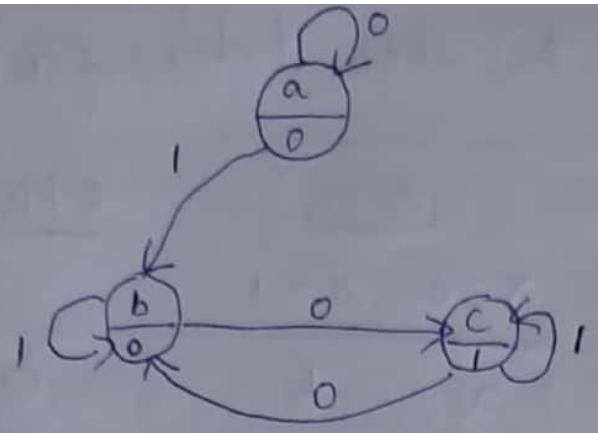
Step2 :- States b and d are Equivalent, replace d by b.

P.S	N.S X=0	N.S X=1	O/P (Z)
a	a	b	0
b	c	b	0
c	b	e	1
e	b	c	1

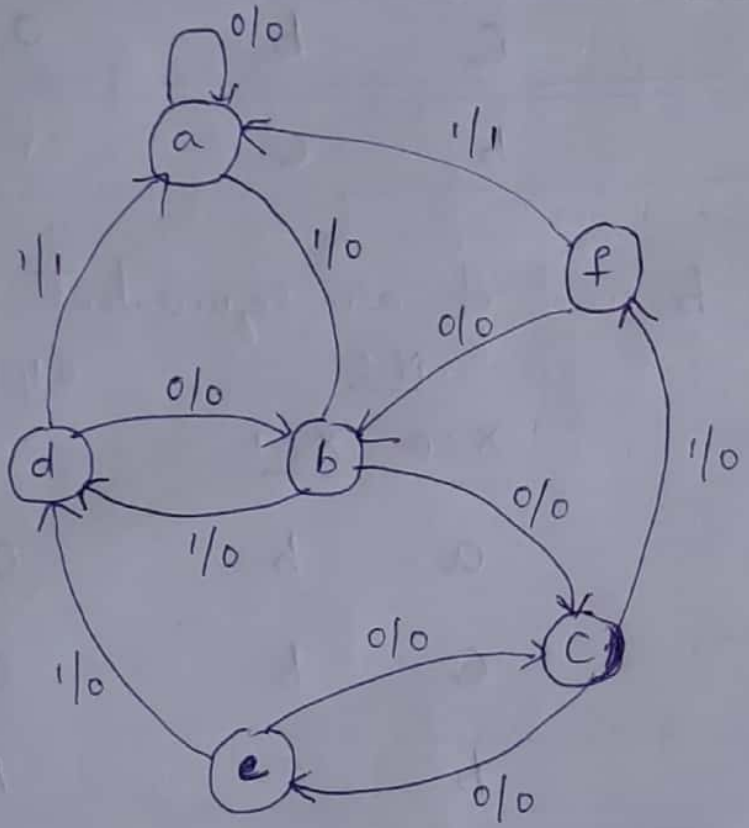
States c & e are Equivalent. Replace e by c & retaining c as it is.

P.S	N.S X=0	N.S X=1	O/P (Z)
a	a	b	0
b	c	b	0
c	b	c	1

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4) Reduce the state transition diagram of Mealy model by row elimination method.



Step 1 :- Draw the state table for the given state diagram

P.S	N.S		output	
	X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	e	f	0	0
d	b	a	0	1
e	c	d	0	0
f	b	a	0	1

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Step2 :- States b & e are Equivalent. Retain b & replace e by b.

P.S	N.S X=0	N.S X=1	O/P(Z) X=0	O/P(Z) X=1
a	a	b	0	0
b	c	d	0	0
c	b	f	0	0
d	b	a	0	1
f	b	a	0	1

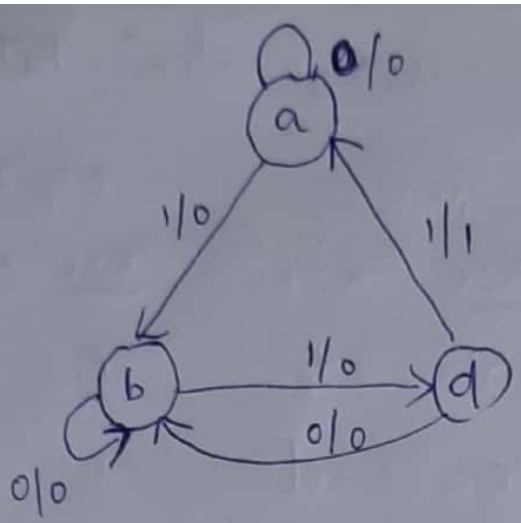
States d & f are Equivalent. Retaining d & replace f by d

P.S	N.S X=0	N.S X=1	O/P(Z) X=0	O/P(Z) X=1
a	a	b	0	0
b	c	d	0	0
c	b	d	0	0
d	b	a	0	1

States b & c are Equivalent. Retaining b & replace c by b

P.S	N.S X=0	N.S X=1	O/P(Z) X=0	O/P(Z) X=1
a	a	b	0	0
b	b	d	0	0
d	b	a	0	1

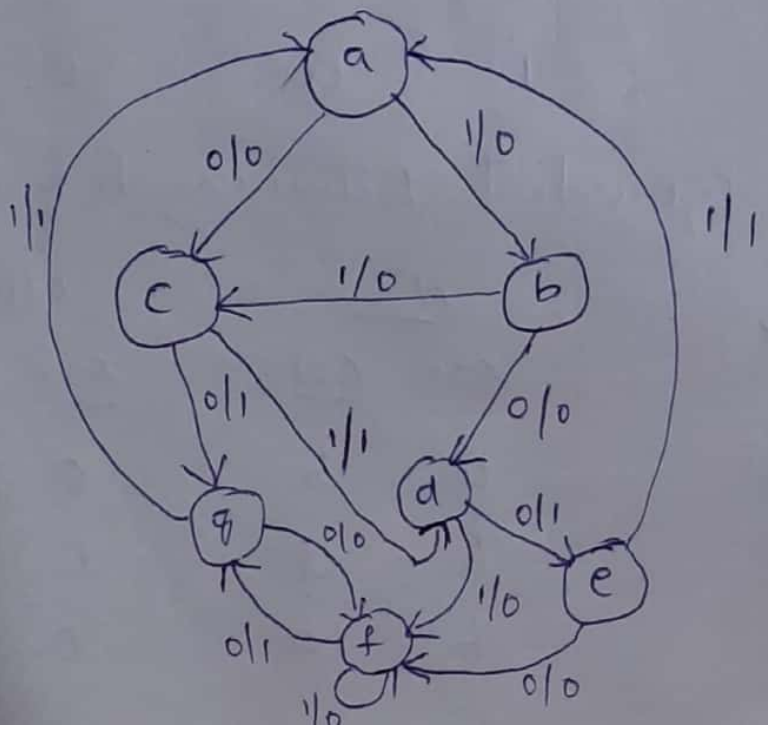
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ii) Implication Table method :- we identify the states, which cannot be equivalent, as their outputs do not match. This we indicate by putting a double-cross in respective cross points.

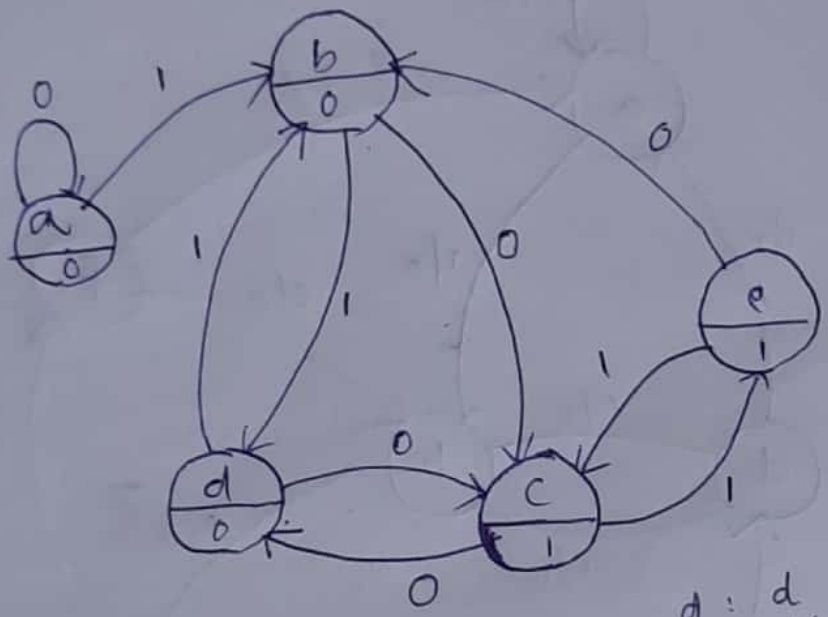
For other cross sections, we write necessary condition for equivalence of intersecting states.

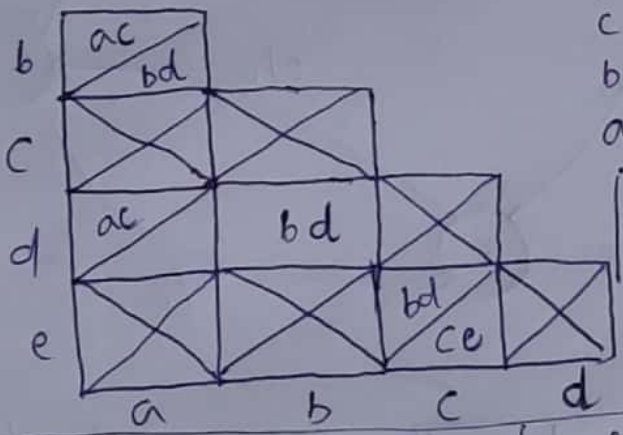
i) Reduce the states by implication table method for the state transition diagram.



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2) Reduce the given State transition diagram by Implication Table method. **05** (15)





d : d
c : d(ce)
b : d(ce)(bd) = (ce)(bd)
a : (ce)(bd) a

$P = (ce)(bd)(a)$

Initial table

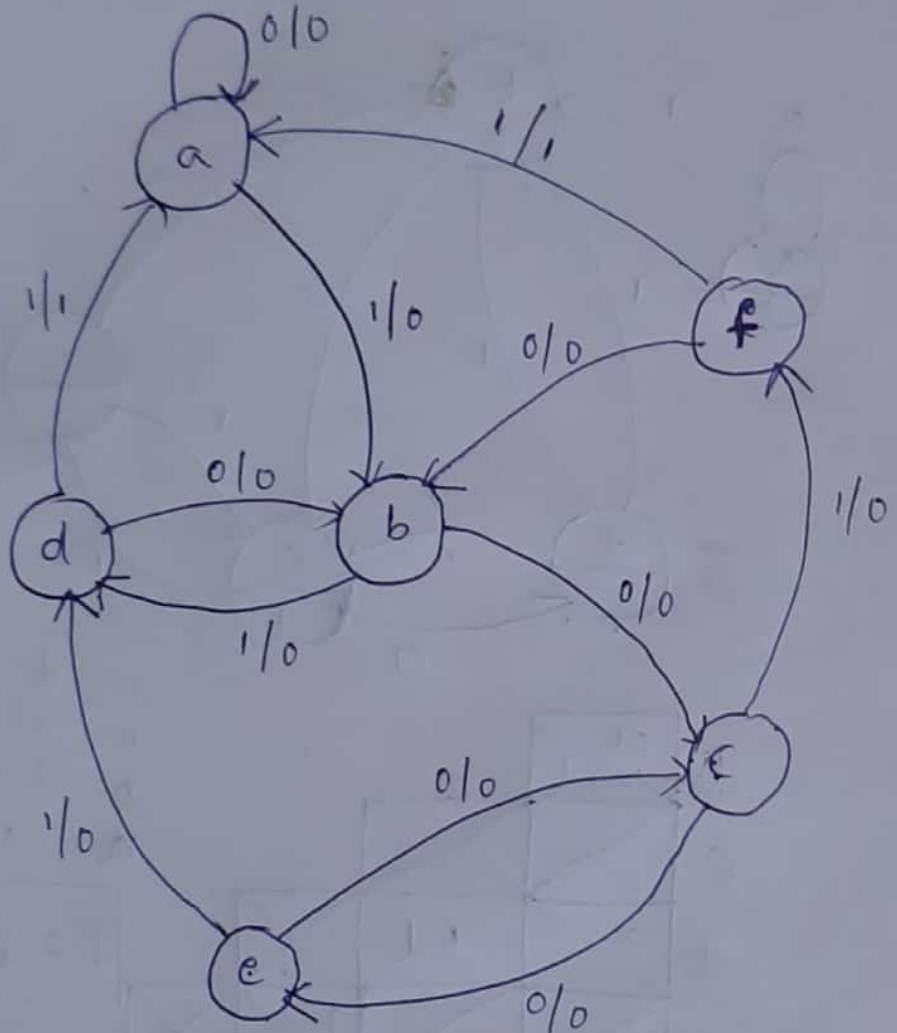
P S	N S X=0	N S X=1	O/P Y
a	a	b	0
b	c	d	0
c	d	e	1
d	c	b	0
e	b	c	1

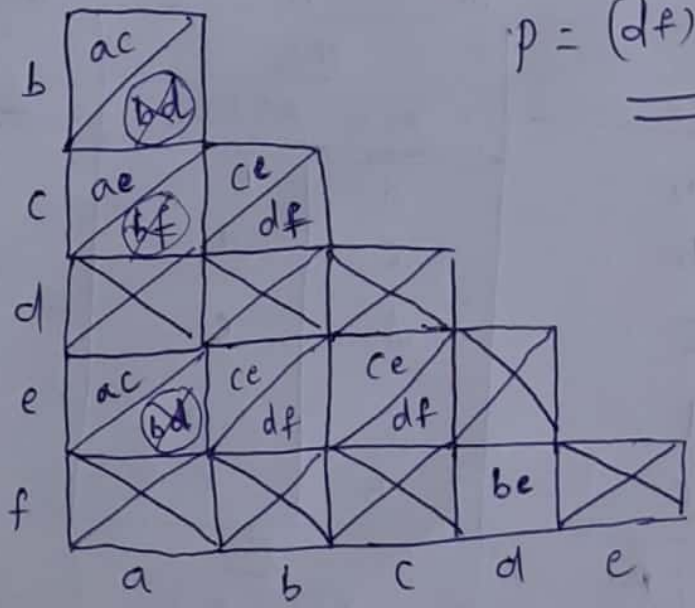
Find table after reducing states

P.S	N.S X=0	N.S X=1	O/P
a	a	b	0
b	c	b	0
c	b	c	1

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3) 06 Reduce the state transition diagram of Mealy model by Implication table method.





$P = (df)(bce)\,a$

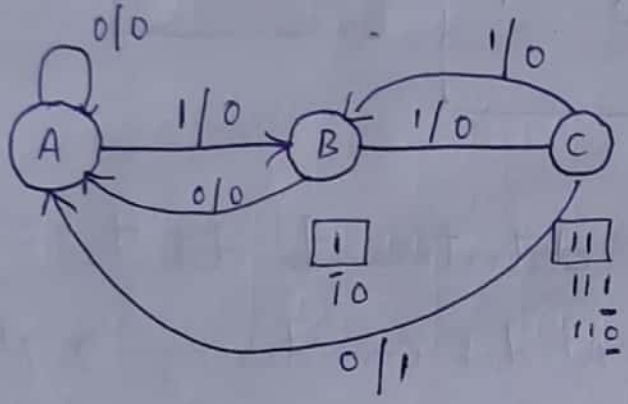
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Write the difference between Synchronous sequential and Asynchronous Sequential circuits. 07 (16)

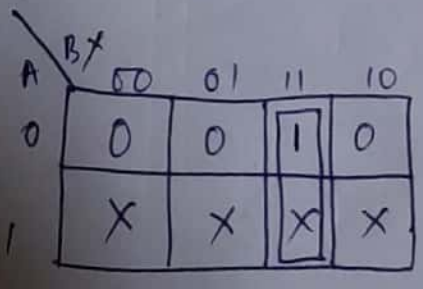
Synchronous Sequential circuit.	Asynchronous Sequential circuit
In Synchronous circuit, Memory elements are clocked flip-flops.	In asynchronous circuit, memory elements are either unclocked flip-flops or time delay elements.
In Synchronous circuit, the change in input signals can affect memory elements upon activation of clock signal.	In asynchronous circuit, change in input signals can affect memory elements at any instant of time.
The maximum operating speed of clock depends on time delays involved.	Because of absence of clock, asynchronous circuits operate faster than sequential circuits
Easier to design.	More difficult to design.

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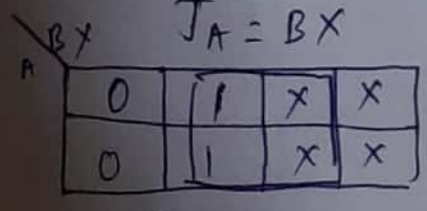
Design a sequence detector to detect the following sequence using JK flip-flops use Mealy machine and the sequence is 110. (1 bit overlapping) 09 (17)



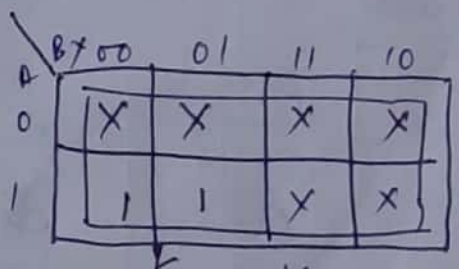
P.S		I/P	N.S		O/P	FF I/P			
A	B	X	A^+	B^+	Y	J_A	K_A	J_B	K_B
0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	0	X	1	X
0	1	0	0	0	0	0	X	X	1
0	1	1	1	0	0	1	X	X	1
1	0	0	0	0	1	X	1	0	X
1	0	1	0	1	0	X	1	1	X



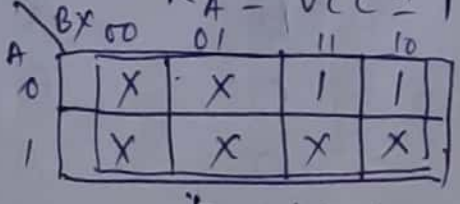
$J_A = BX$



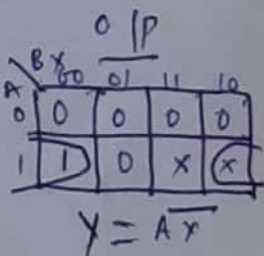
$J_B = X$



$K_A = Vcc = 1$



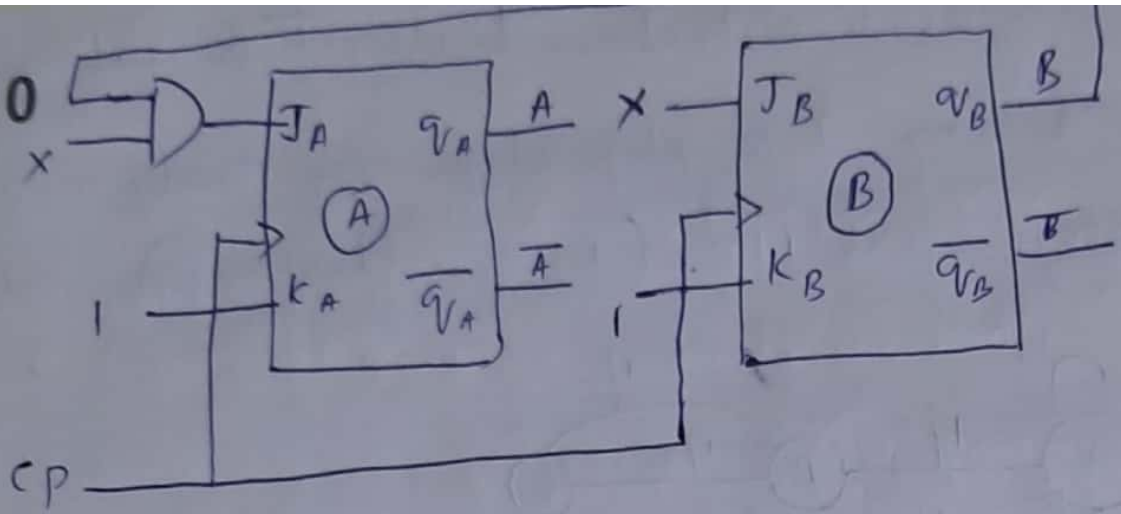
$K_B = 1 = Vcc$



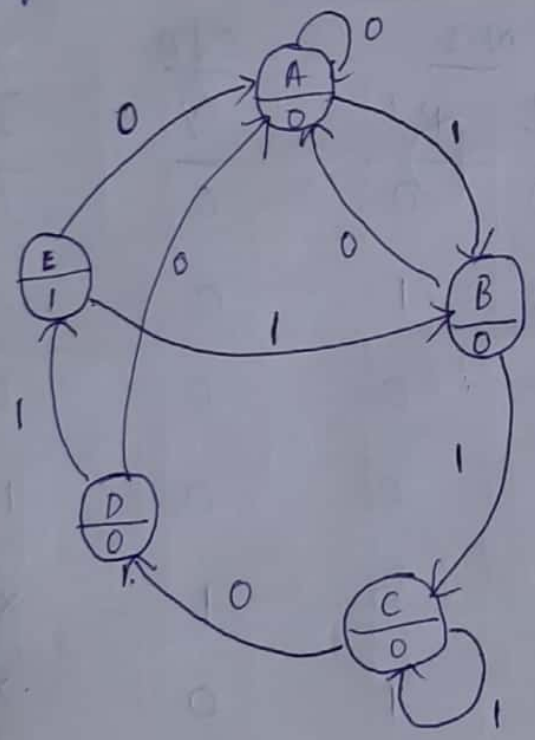
$Y = A\bar{X}$

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10



Design a sequence detector to detect the sequence 1101 using Moore Model, Considering D flip-flop.

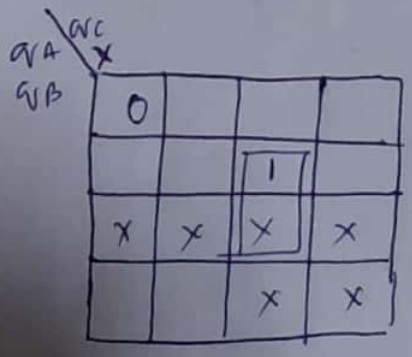


State table

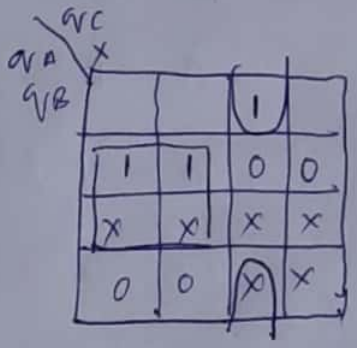
P.S	N.S		O/P
	x=0	x=1	Y
A	A	B	0
B	A	C	0
C	D	C	0
D	A	E	0
E	A	B	1

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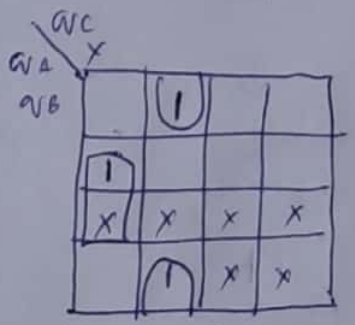
P S			I/P	N S			O/P	FF I/P		
Q_A	q_B	q_C	X	q_A^+	q_B^+	q_C^+	Y	D_A	D_B	D_C
0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0
0	0	1	1	0	1	0	0	0	1	0
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	0	1	0	0	0	1	0
0	1	1	0	0	0	0	0	0	0	0
0	1	1	1	1	0	0	0	1	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	0	1	0	0	1	1	0	0	1



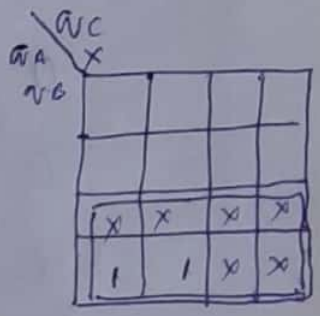
$D_A = q_B q_C x$



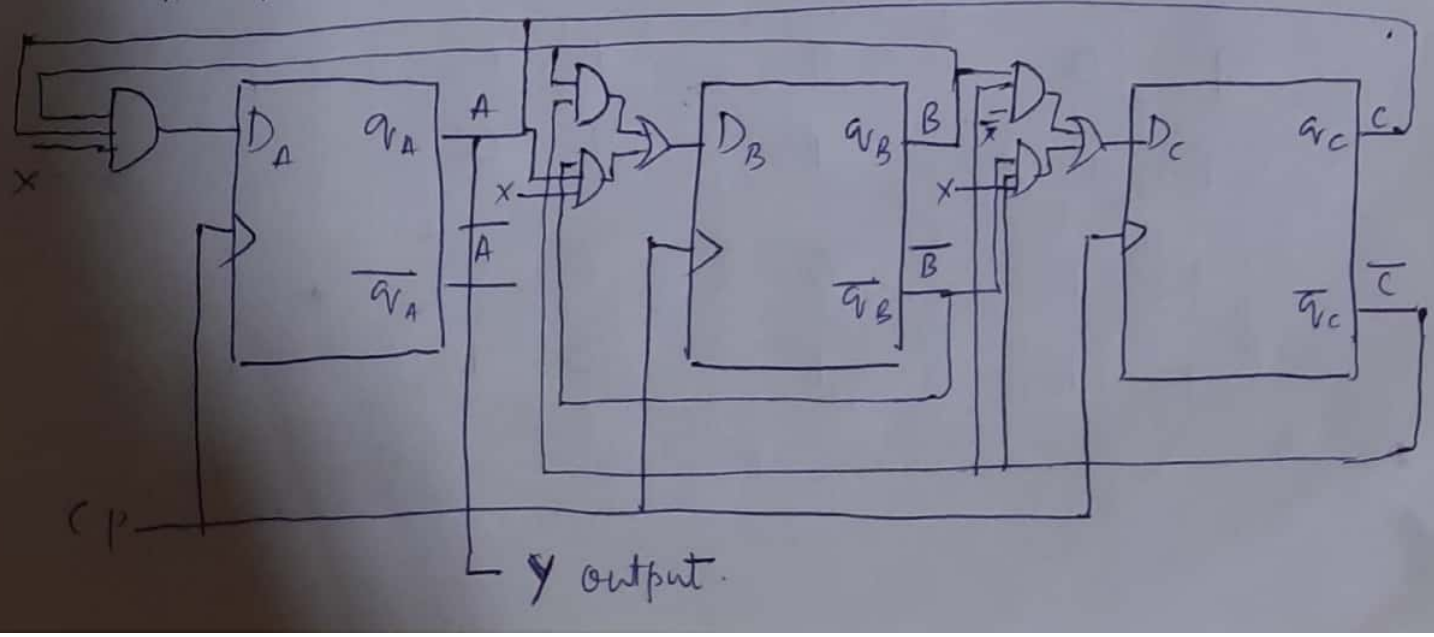
$D_B = q_B \bar{q}_C + \bar{q}_B q_C x$



$D_C = q_B \bar{q}_C \bar{x} + \bar{q}_B \bar{q}_C x$

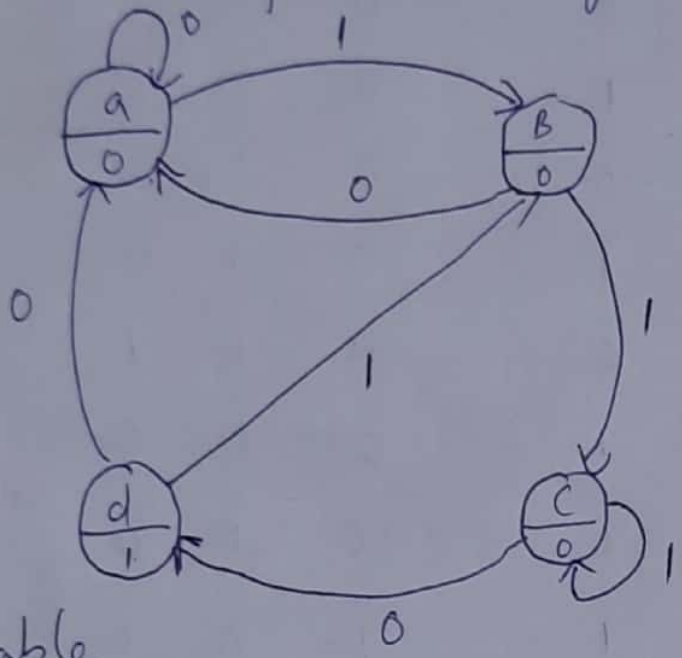


$Y = q_A$



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Design the Moore type sequential circuit to implement sequence detector represented by state diagram.



P.S	I/P	N.S	Y
a	0	a	0
a	1	b	0
b	0	a	0
b	1	c	0
c	0	d	0
c	1	c	0
d	0	a	1
d	1	b	1

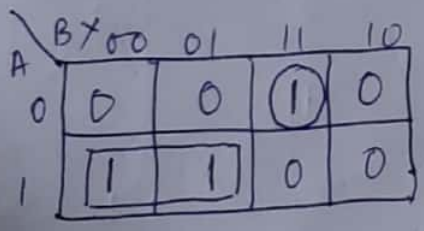
O/P

State table

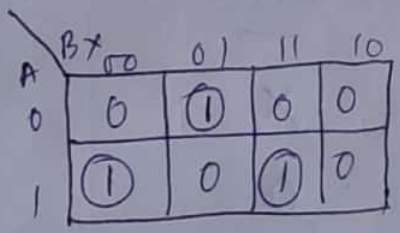
P.S		I/P	N.S		O/P
A	B	X	A^+	B^+	Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	1	1	0
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	0	1	1

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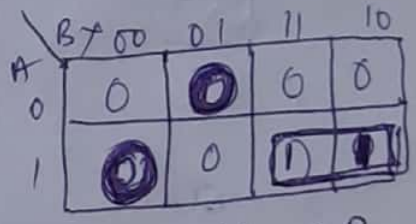
P.S		I/P	N.S		O/P	FF I/P	
A	B	X	A^+	B^+	Y	D_A	D_B
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	0	0	0	0
0	1	1	1	0	0	1	0
1	0	0	1	1	0	1	1
1	0	1	1	0	0	1	0
1	1	0	0	0	1	0	0
1	1	1	0	1	1	0	1



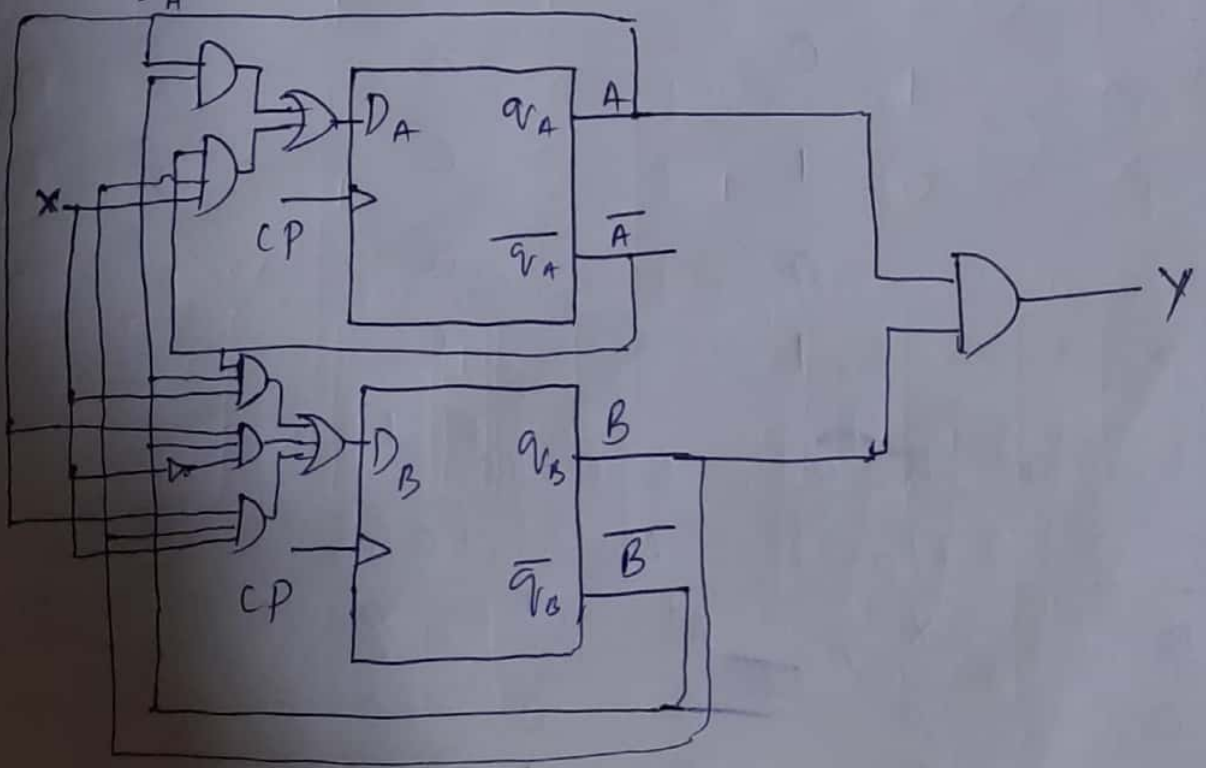
$D_A = A\bar{B} + \bar{A}BX$



$D_B = \bar{A}\bar{B}X + A\bar{B}\bar{X} + ABX$

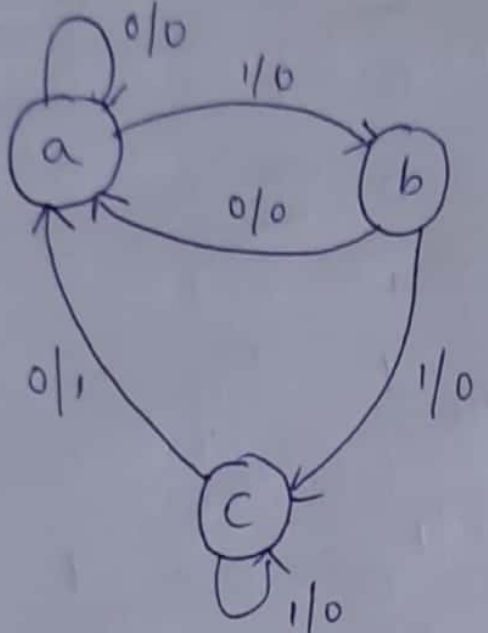


$Y = AB$



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Design the mealy type sequential circuit to implement sequential detector represented by state transition diagram:



State table

P.S	I/P	N.S	O/P
	X		Y
a	0	a	0
a	1	b	0
b	0	a	0
b	1	c	0
c	0	a	1
c	1	c	0

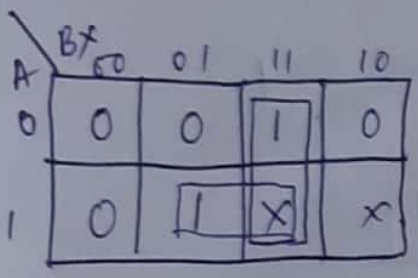
assigning a = 00, b = 01, & c = 10

P.S		I/P	N.S		O/P	FF I/P	
A	B	X	A^+	B^+	Y	D_A	D_B
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1

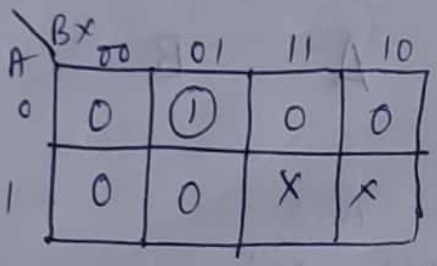
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(20)

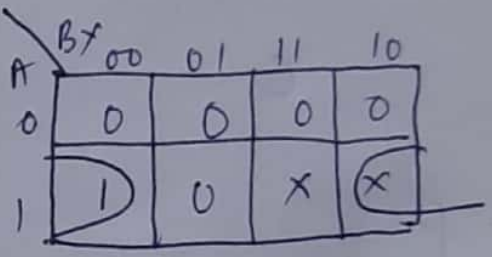
0	1	0	0	0	0	0
0	1	1	1	0	0	1	0
1	0	0	0	0	1	0	0
1	0	1	1	0	0	1	0



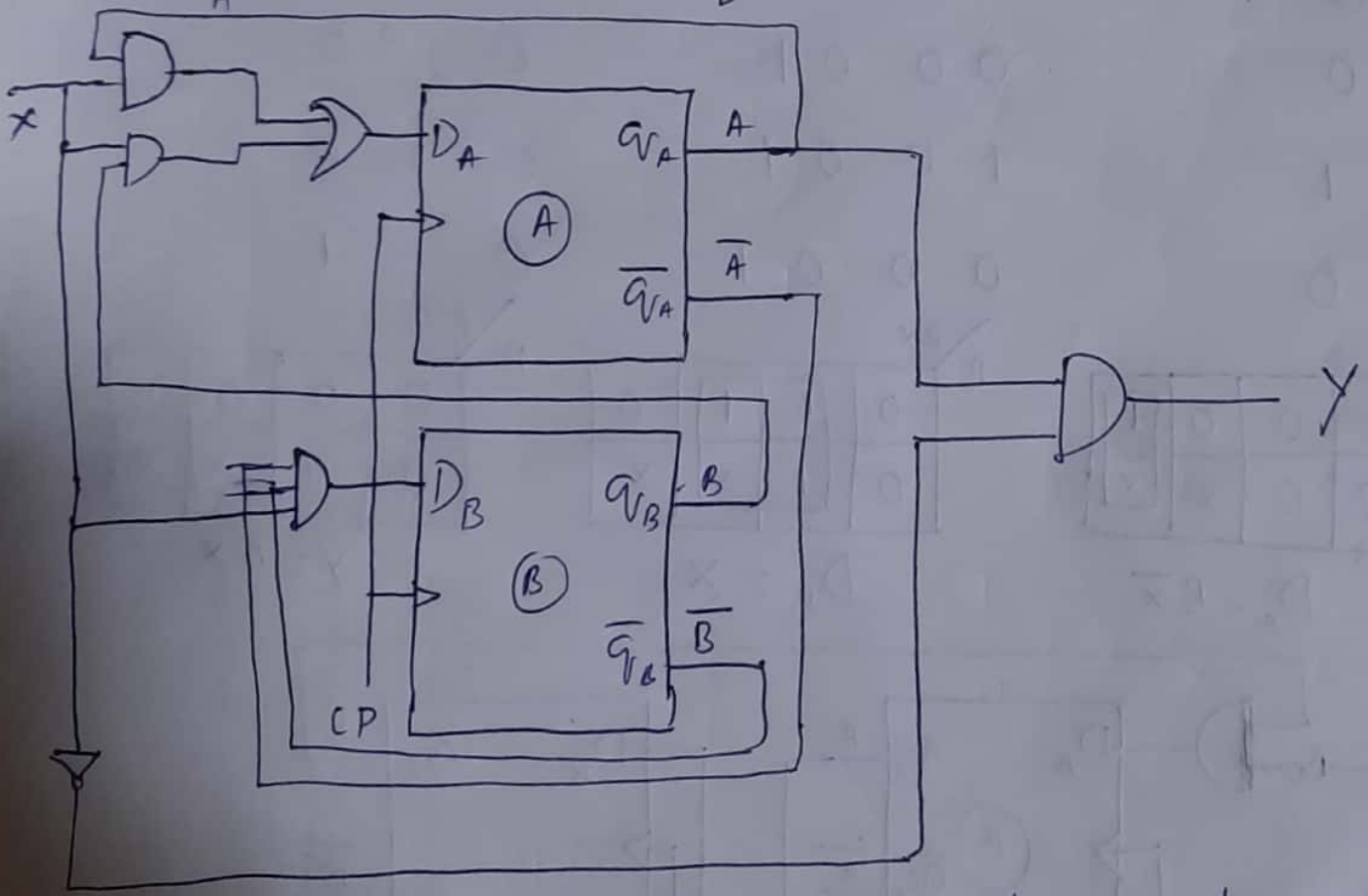
$D_A = Bx + Ax$



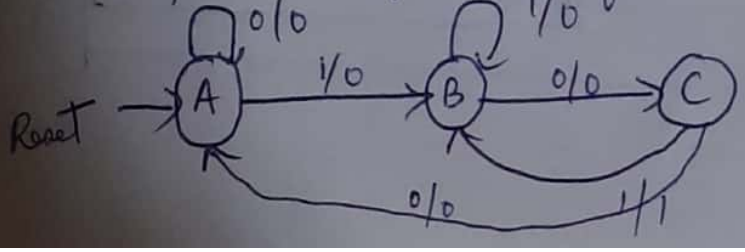
$D_B = \bar{A}\bar{B}x$



$Y = A\bar{x}$



Design a mealy type sequence detector to detect a serial input sequence of 101

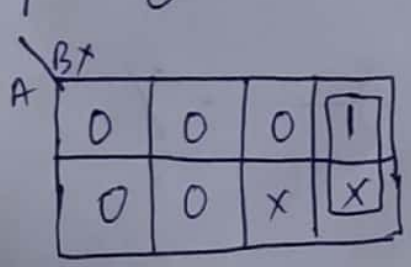


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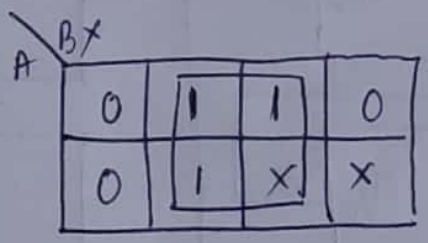
14 State table

P.S	N.S		O/P	
	X=0	X=1	X=0	X=1
A	A	B	0	0
B	C	B	0	0
C	A	B	0	1

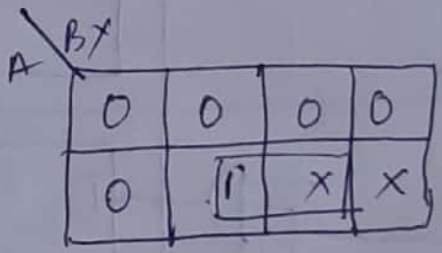
P.S	N.S		O/P	
A B	X=0 A^+	X=1 B^+	X=0	X=1
0 0	0	1	0	0
0 1	1	0	0	0
1 0	0	0	0	1



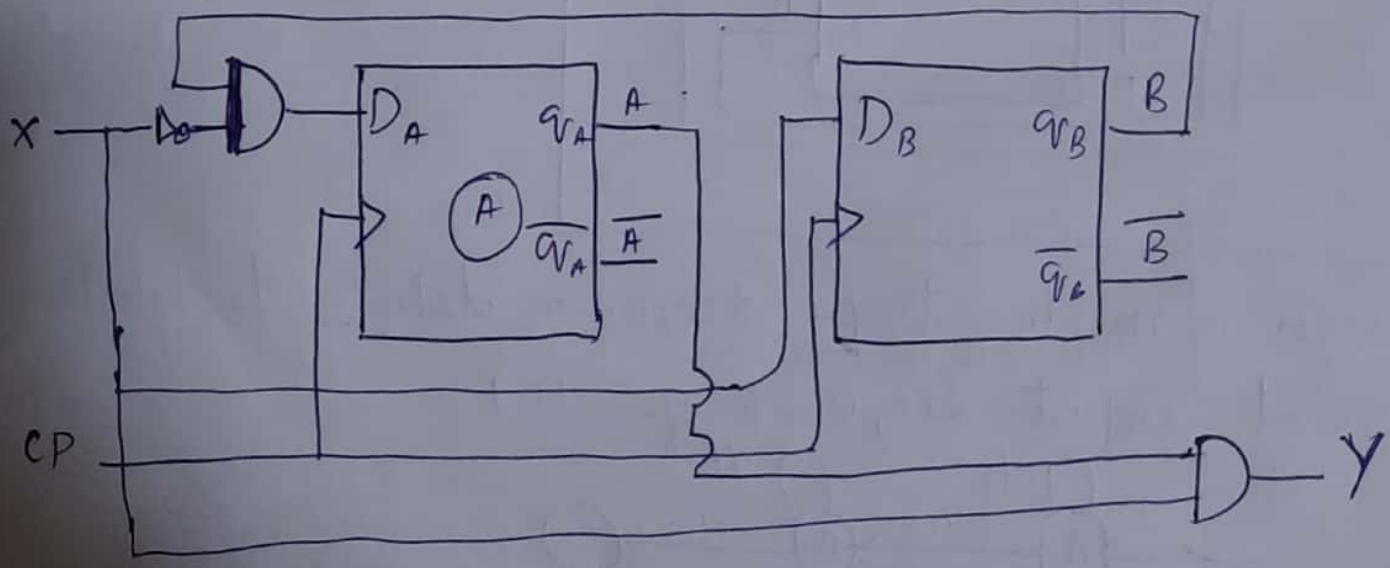
$D_A = B\bar{X}$



$D_B = X$



$Y = AX$



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An asynchronous sequential circuit is described by the following excitation and output function. (21)

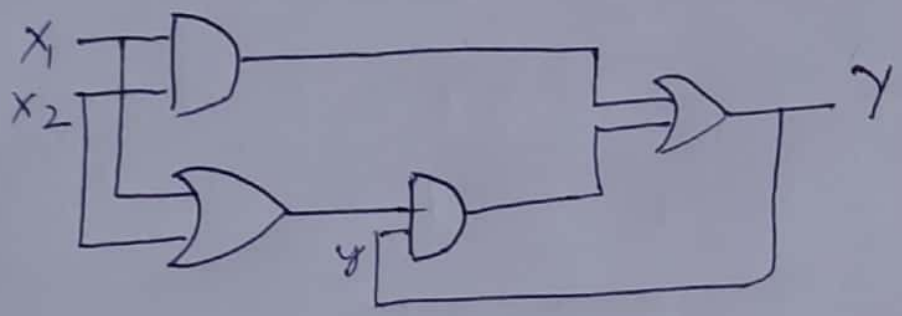
$$Y = x_1 x_2 + (x_1 + x_2) y$$

a) Draw the logic diagram of the circuit.

b) Derive the transition table and output map.

c) Describe the behavior of the circuit.

a)



P.S			N.S			Stable state	output(Y)
y	x_1	x_2	y	x_1	x_2	Yes/No	
0	0	0	0	0	0	Yes	0
0	0	1	0	0	1	Yes	0
0	1	0	0	1	0	Yes	0
0	1	1	1	1	1	No	1
1	0	0	0	0	0	No	0
1	0	1	1	0	1	Yes	1
1	1	0	1	1	0	Yes	1
1	1	1	1	1	1	Yes	1

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